# Audio Equipment and Systems

Bachelor's Degree in Audiovisual Systems Engineering

## Sound and Acoustics

Engineering School
Department of Information
and Communication Technologies

Universitat Pompeu Fabra Barcelona

Adrià Garcia Riber adrian.garcia@upf.edu
Xavier Lizarraga Seijas xavier.lizarraga@upf.edu

# Sound and Acoustics

- Sound generation and propagation
- Distance to source influence. The inverse square law
- Directivity factor and directivity index
- Sound field definitions
- Human ear and equal loudness
- Masking and critical band
- Phase and correlation
- Precedence effect and comb filter
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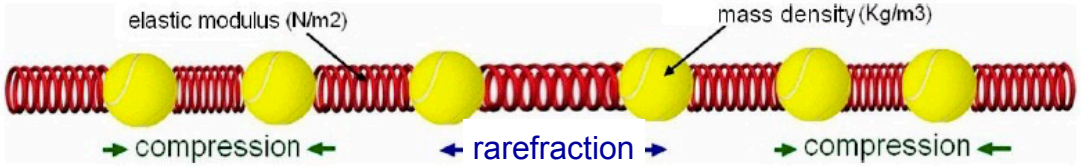
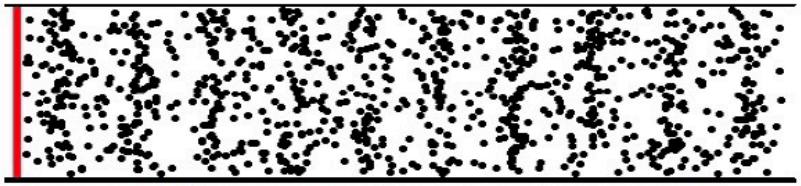
- Phase relationship and coherence
- Electro-acoustic duality
- Decibels
- Measurement scales
- Peak and RMS levels. Crest factor
- Goniometer and Lissajous curves
- Cross-correlation
- RTA, Magnitude and phase analysis
- Absorption, reflection, refraction, and diffraction
- Reflection vs Diffusion
- Acoustic Isolation. Soundproofing
- Room modes
- Acoustic conditioning. Reflection design
- Acoustic materials
- Variable acoustics

# Sound generation and propagation

- propagation on solid / liquid / gaseous media → all of them with a certain *elasticity*
- sound propagation speed




- on a gaseous medium such as air, propagation is longitudinal



$$\text{sound speed (m/s)} = \sqrt{\frac{\text{stiffness (Young's modulus, N/m2)}}{\text{mass density (Kg/m3)}}}$$

Stiffness dominates the effect

- propagation speed in a gas depens on
  - gas density → $v_{dry\ air} < v_{moist\ air}$ → Heavier nitrogen and oxygen molecules are replaced by lighter (lower molecular weight) water vapor molecules, reducing the overall density of the air
  - temperature → $v \approx 331+0{,}6T[ºC]$ (m/s)

| material | characteristics | Young's modulus (N/m2) | density (Kg/m3) | sound speed (m/s) |
|---|---|---|---|---|
| steel | | 2,10x10E11 | 7800 | 5189 |
| glass | | 6,00x10E10 | 2400 | 5000 |
| concrete | | 3,00x10E10 | 2400 | 2536 |
| lead | | 1,70x10E10 | 11400 | 1221 |
| beech wood | along grain | 1,40x10E10 | 680 | 4537 |
| | across grain | 8,80x10E8 | 680 | 1138 |
| water | | 2,30x10E9 | 1000 | 1517 |
| air | at 20°C | 1,43x10E5 | 1.21 | 344 |

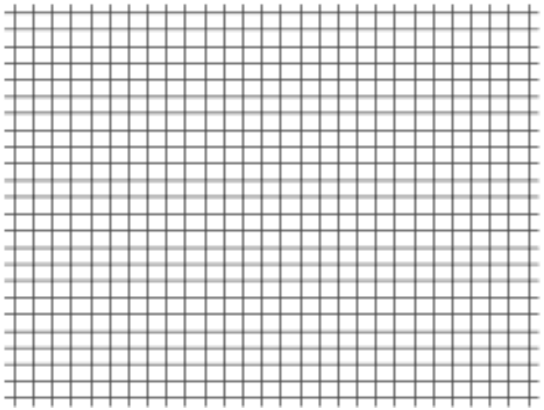
Animations: https://www.acs.psu.edu/drussell/demos.html

Video: https://youtu.be/nl2q_g_Mors?feature=shared&t=586

# Sound generation and propagation

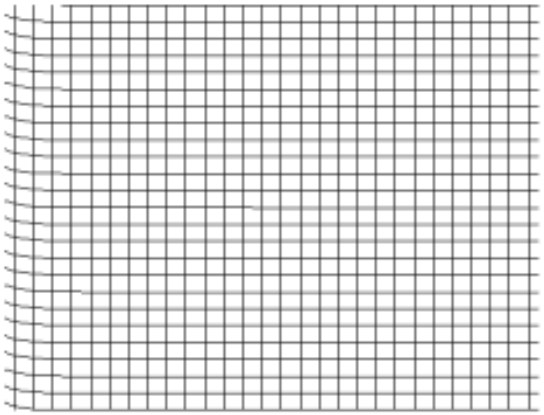
Longitudinal propagation
Example. Sound waves on air

https://en.wikipedia.org/wiki/Longitudinal_wave



Transversal propagation
Ex. Membrane of a drum
(elastic solids)

https://en.wikipedia.org/wiki/Transverse_wave



Case study: Marimba

https://www.lafavre.us/tuning-marimba.htm

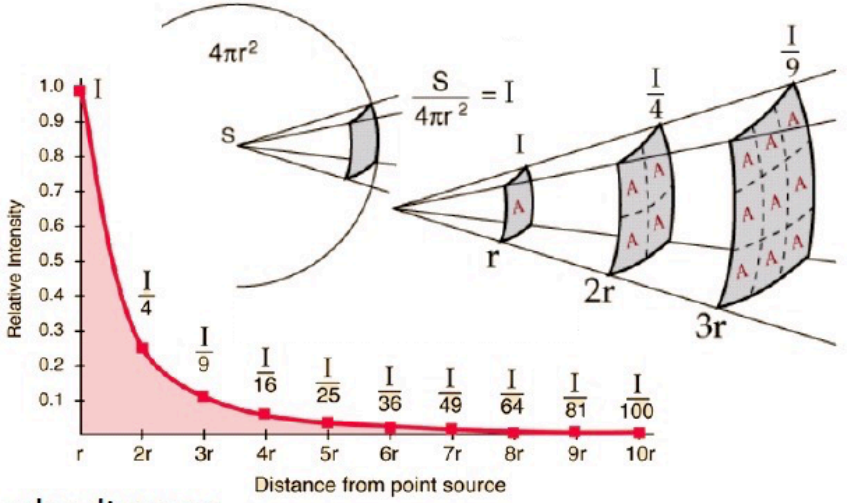
More basics on Musical Acoustics: http://www.phys.unsw.edu.au/jw/basics.html#string

# Distance to source influence

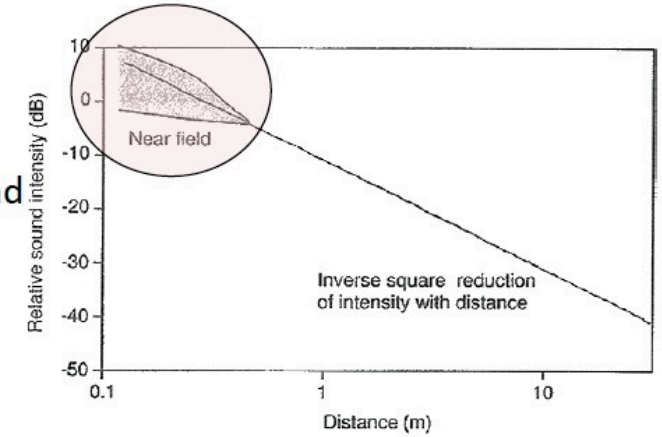
- inverse square law (ISL)

$$I = \frac{Q \cdot W}{4\pi r^2}$$

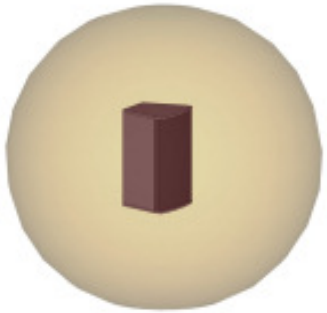
**omnidirectional radiation implies $S=4\pi R^2$**



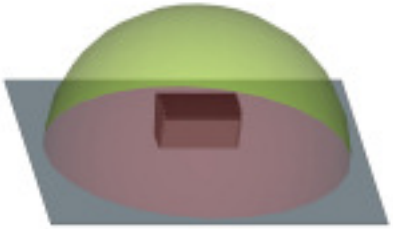

- for omnidirectional radiation, by doubling the distance from the point source sound power is re-distributed over a surface 4 times larger (S → 4S)
- the law becomes unpredictable for distances below critical (near filed behaviour)
- doubling the distance reduces sound Intensity by 4 and sound Pressure by 2 → **-6dB SIL,SPL/double R**
- doubling acoustic power implies multipling sound Intensity by 2 and sound Pressure by 1,41 → **+3dB SIL,SPL/double W**



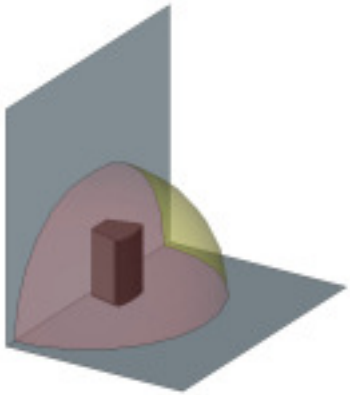
# Directivity factor and directivity index



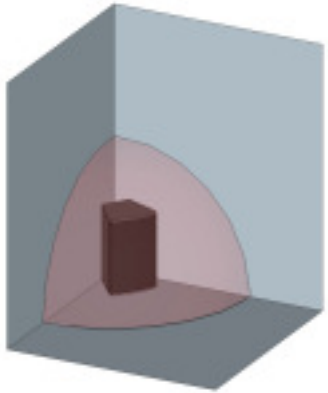
Omnidirectional
Spherical
Q=1 / DI=0dB



One surface
1/2 sphere
Q=2 / DI=+3dB



Two surfaces
1/4 sphere
Q=4 / DI=+6dB



Three surfaces
1/8 sphere
Q=8 / DI=+9dB

+3dB SIL, SPL every time we double Q

# Spherical vs cylindrical radiation

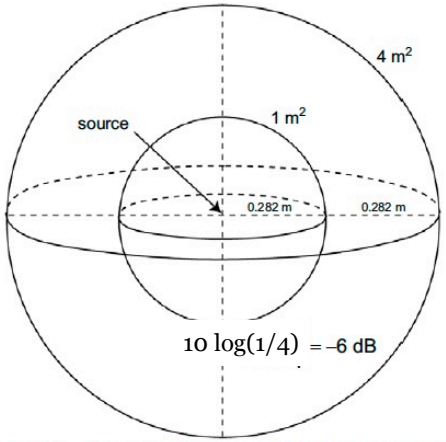



**Figure 2-22.** When the distance to the source is doubled, the radiated sound energy will be spread over 4 times the area. Both $L_I$ and $L_P$ will drop by 6 dB. Courtesy Syn-Aud-Con.

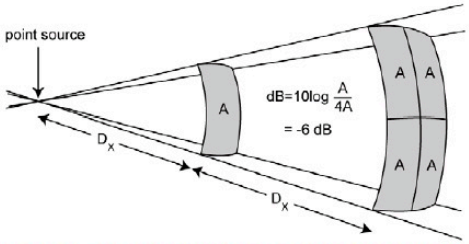



**Figure 2-23.** The ISL is also true for directional devices in their far field (remote locations from the device). Courtesy Syn-Aud-Con.

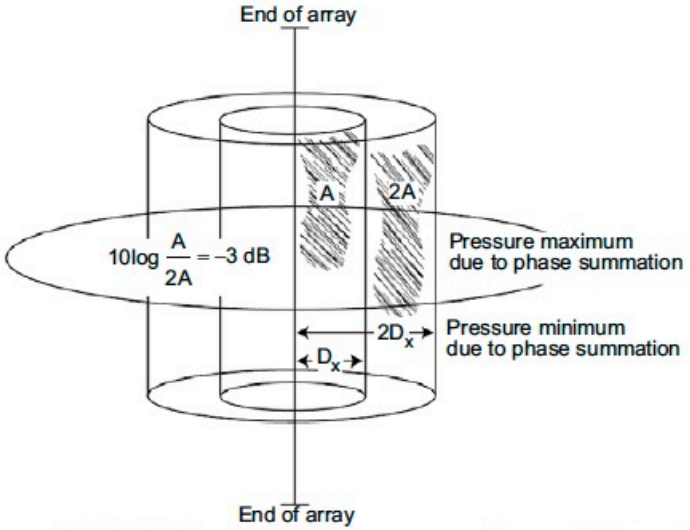



**Figure 2-25.** The finite line array has gained wide acceptance among system designers, allowing wide audience coverage with minimal energy radiation to room surfaces. Courtesy Syn-Aud-Con.

http://www.sengpielaudio.com/calculator-distance.htm

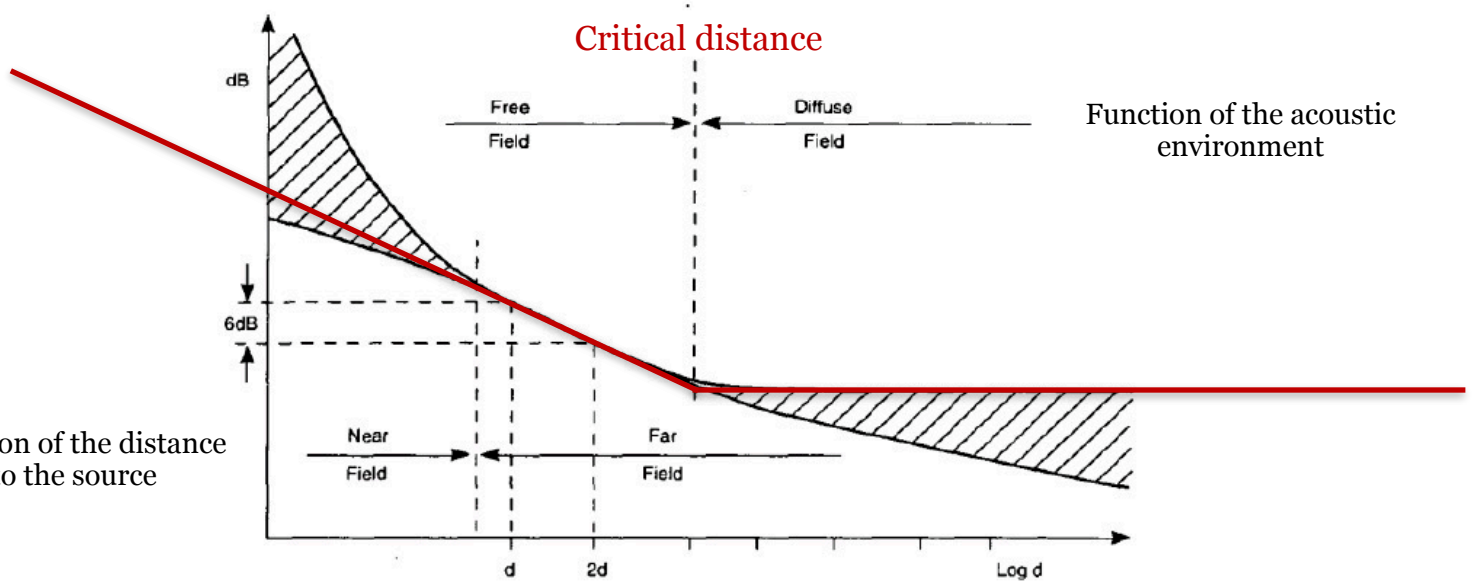
BALLOU, Glen.
*Handbook for sound engineers.*
Taylor & Francis, 2013, pp 37-38.

# Sound field definitions

Critical distance (indoor): is the distance from a sound source inside a room or venue at which the direct sound and the reverberant sound are equal in level.

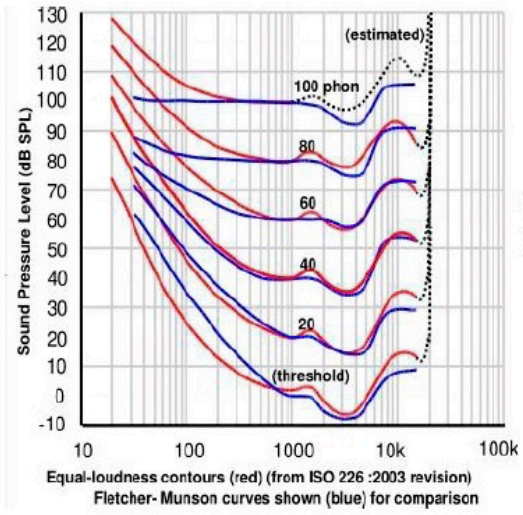
- **free field** => We can capture the direct sound from the source.
  - idealized sound propagation with no reflections (anechoic chamber condition)

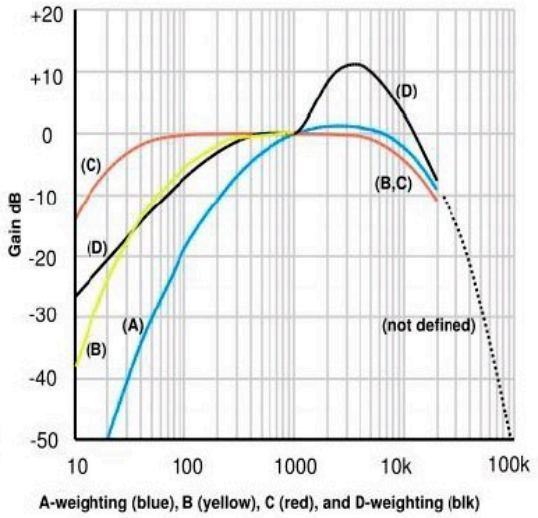
    Indoor => reflexions but -6dB/dd at Near to Mid distances from the source (Example: recording studios)
  - the inverse square law applies (-6dB for every doubling distance)
- **difuse field** => We can capture the influence of the environment.
  - sound arrives from all directions with equal magnitude and probability (delayed and attenuated reflections, mixed-phase systems)



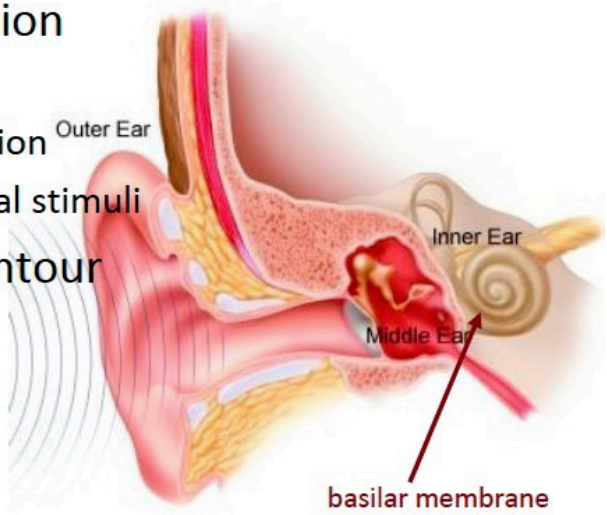
# Human ear and equal loudness

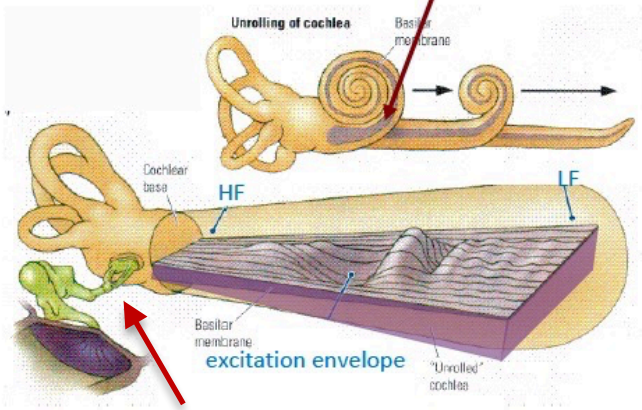
- anatomy of hearing system, responsible of perception
  - outer ear → source location, 4KHz ressonance zone
  - middle ear → impedance matching (lever effect) & protection
  - inner ear (cochlea) → spectral analysis, hair cells & neuronal stimuli
- hearing threshold & isophonic (equal-loudness) contour
- dBA/B/C/D ponderation



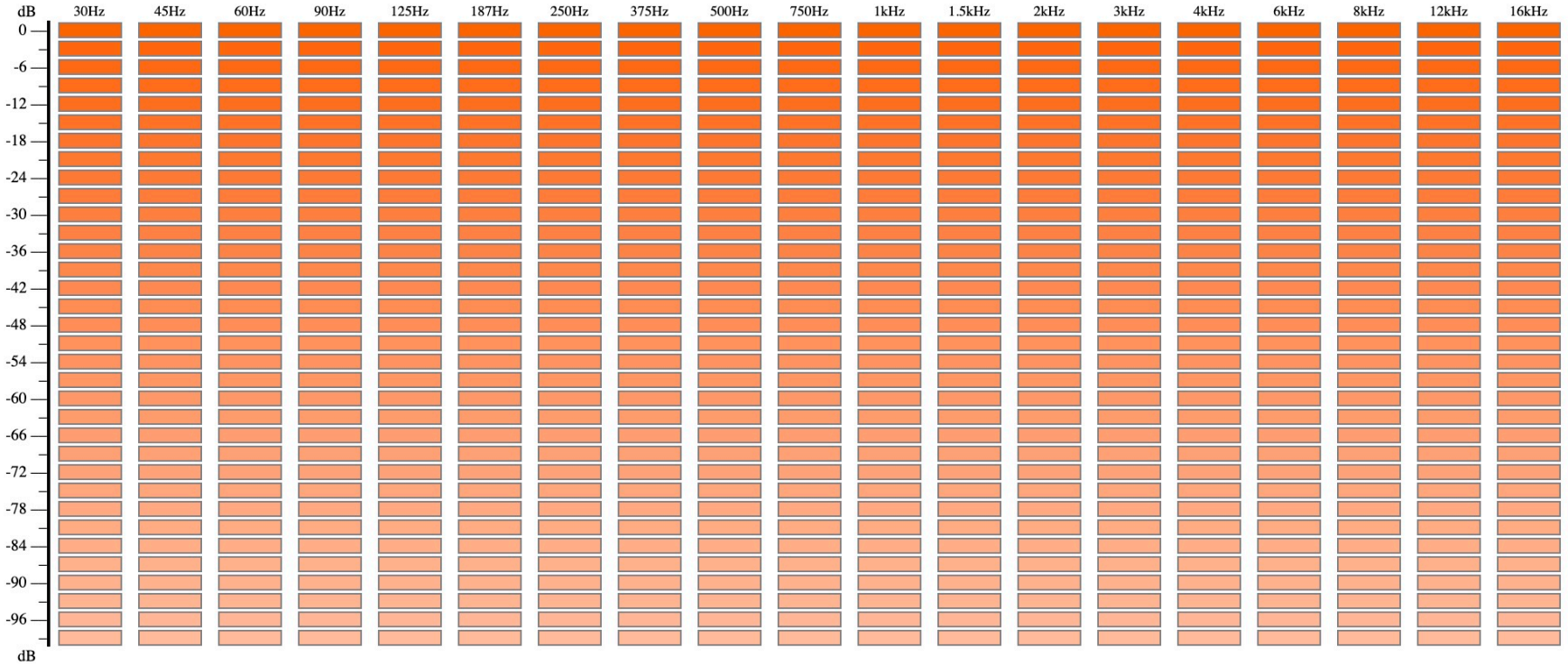

- doubling *sonority perception* implies +10dB SPL




Hammer /anvil /stirrup

# Equal-loudness test

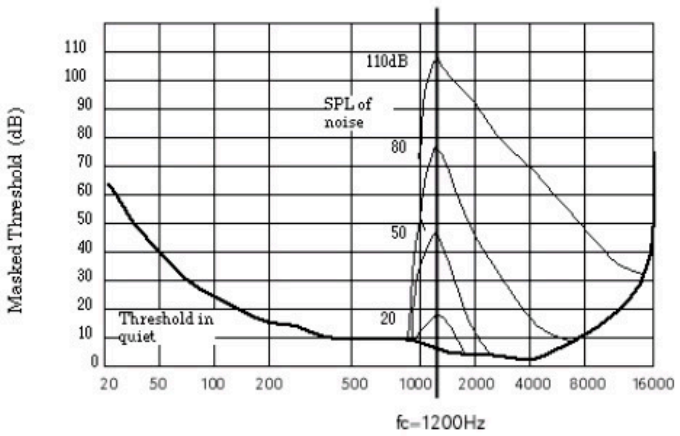


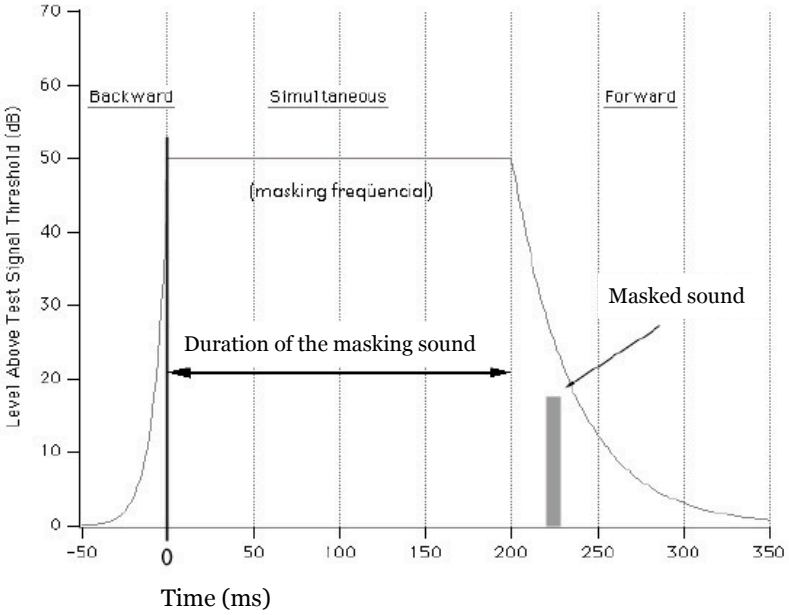

Hearing test on-line: sensitivity, equal loudness contours and audiometry
University of New South Whales. School of Physics.
Sidney, Australia

European Agency for Safety and Health at Work: https://oshwiki.osha.europa.eu/en/themes/noise
(Occupational Safety and Health encyclopedia)

# Masking and critical band

- sound masking
  - frequency masking (simultaneous)
  - temporal masking (non-simultaneous)



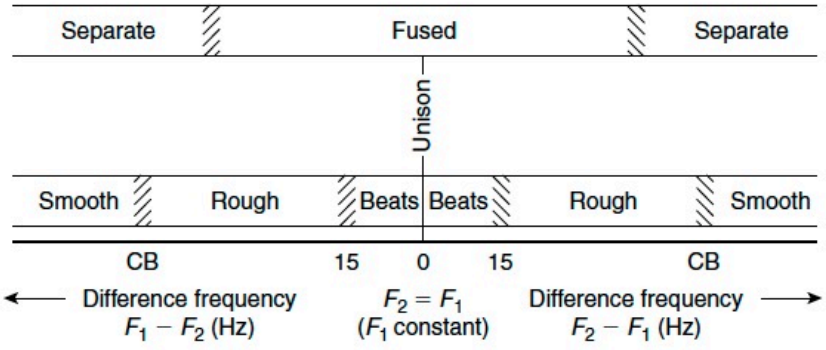



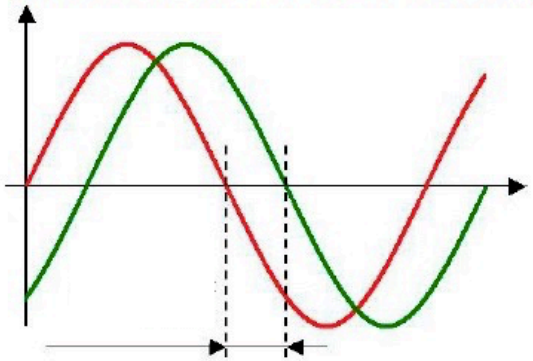

**FIGURE 2.6**
An illustration of the perceptual changes which occur when a pure tone fixed at frequency $F_1$ is heard combined with a pure tone of variable frequency $F_2$.

SELF, Douglas. *Audio engineering explained*. Routledge, 2012, p35.

# Phase and Correlation

- phase differences among sources

$\lambda = c/f$ $f = 1/T$

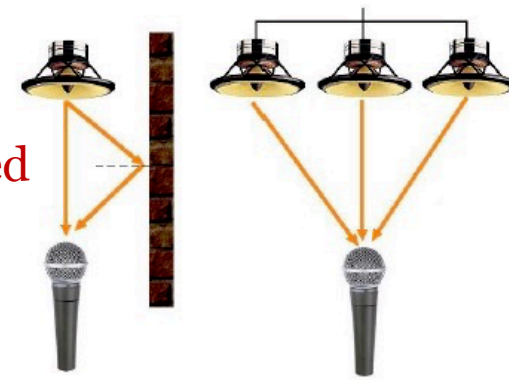


Phase shift

- Correlated signals (coherent)
  - Phase shift = 0º => Perfect addition
  - Phase shift = 180º => Perfect cancellation
  - Any constant phase shift
- Not correlated signals (not coherent)
  - Different signals, not related phase

- addition of correlated (same source & phase relation) vs not correlated sources (different sources / independent phase)

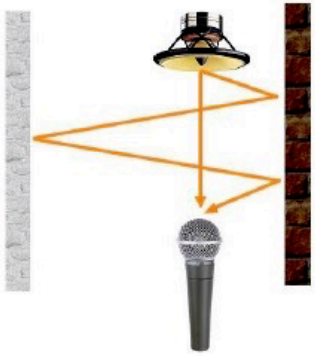
Correlated



Early reflections



Clusters

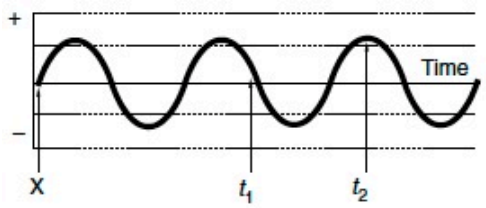


Reverberation

Different sources

Uncorrelated

# Precedence effect and comb filter




Wave leaves both speakers at time X, and arrives at listener at times $t_1$ and $t_2$ respectively

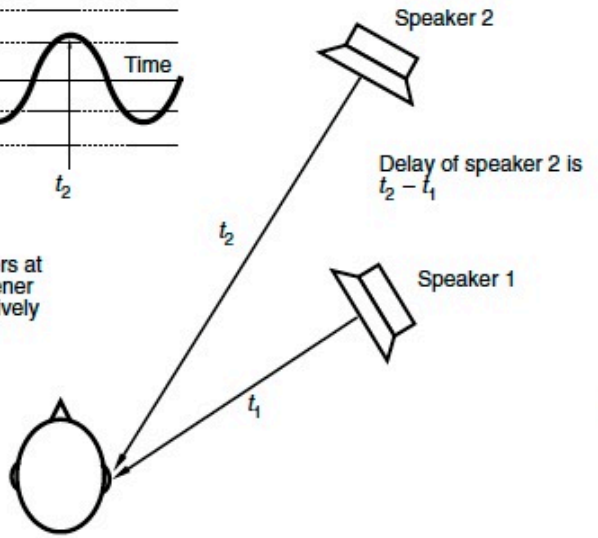



**FIGURE 1.8**
*If the two loudspeakers in the drawing emit the same wave at the same time, the phase difference between the waves at the listener's ear will be directly related to the delay $t_1 - t_2$.*

$$c = \frac{d}{t} \rightarrow t_i = \frac{d_i}{c} \rightarrow \tau = \Delta t = (t_2 - t_1) = \frac{d_2 - d_1}{c}$$

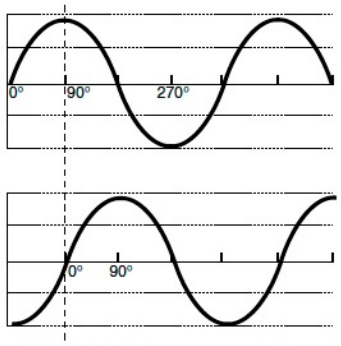
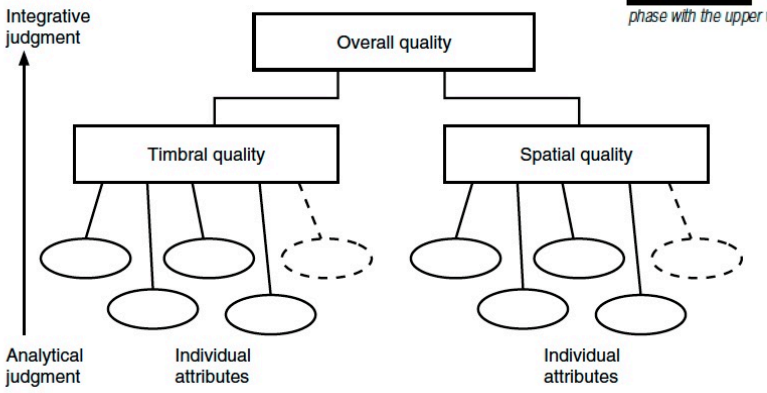
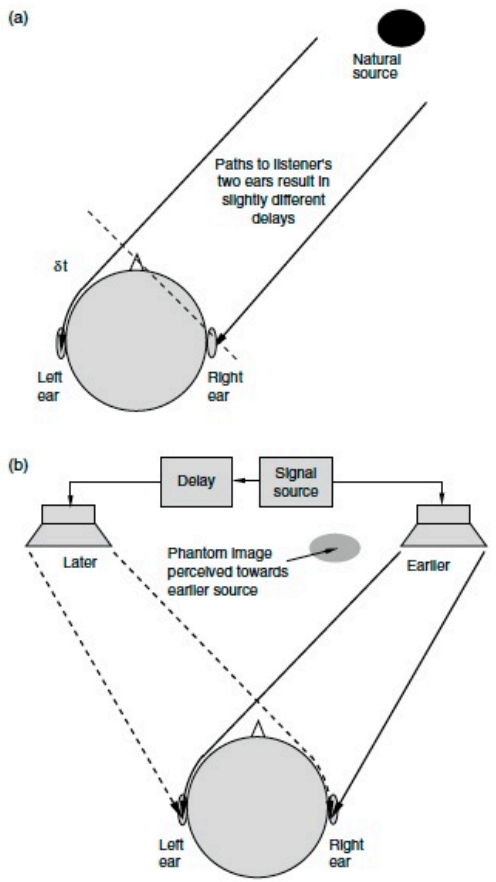



**FIGURE 1.10** *The lower wave is 90° out of phase with the upper wave.*

**FIGURE 18.3**
*Overall sound quality can be subdivided into timbral and spatial quality domains. Each of these consists of contributions from a number of relevant low-level attributes.*



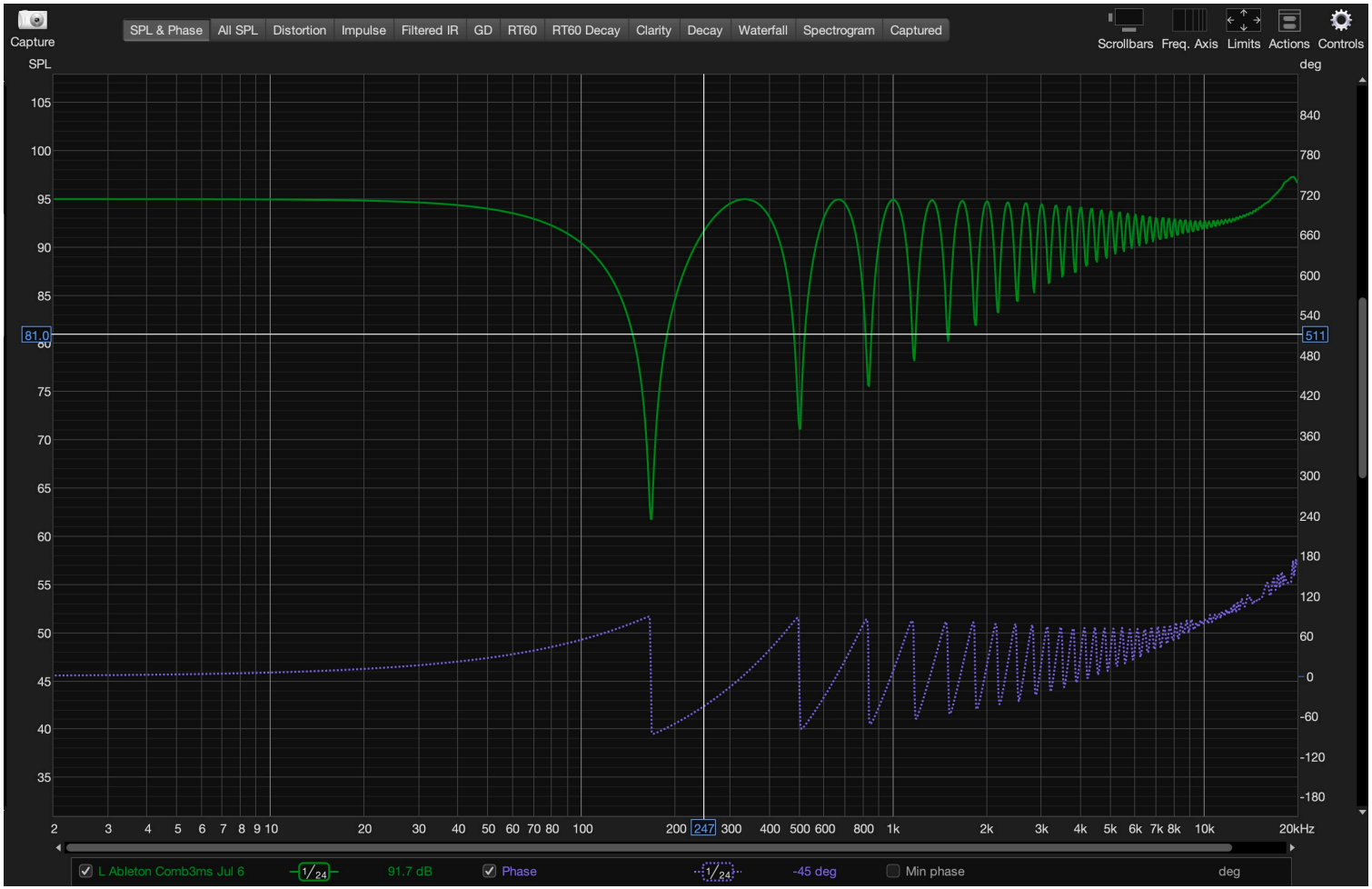

Binaural vs stereophonic




ITD (Interaural Time Difference)
ILD (Interaural Level Difference)

RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012, pp 11, 12, 565 & 476.

# Comb filtered response

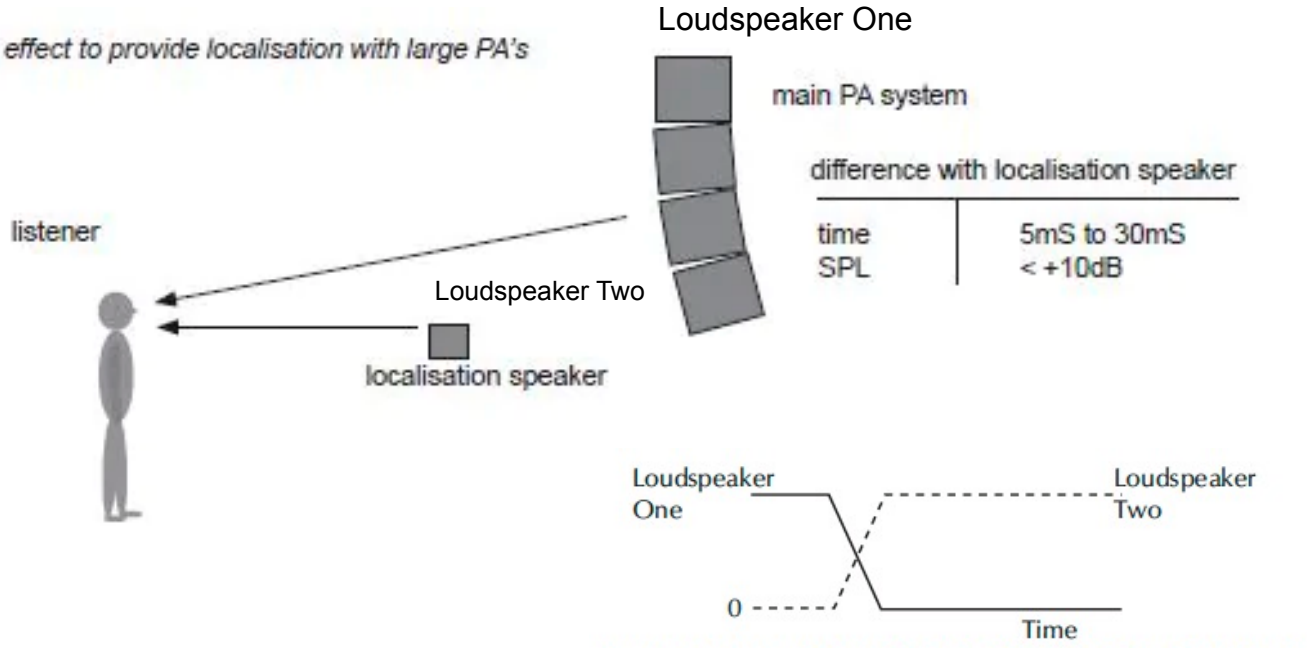



Comb filter corresponding to a 3ms delay.

# Hass, Franssen, and Cocktail Party

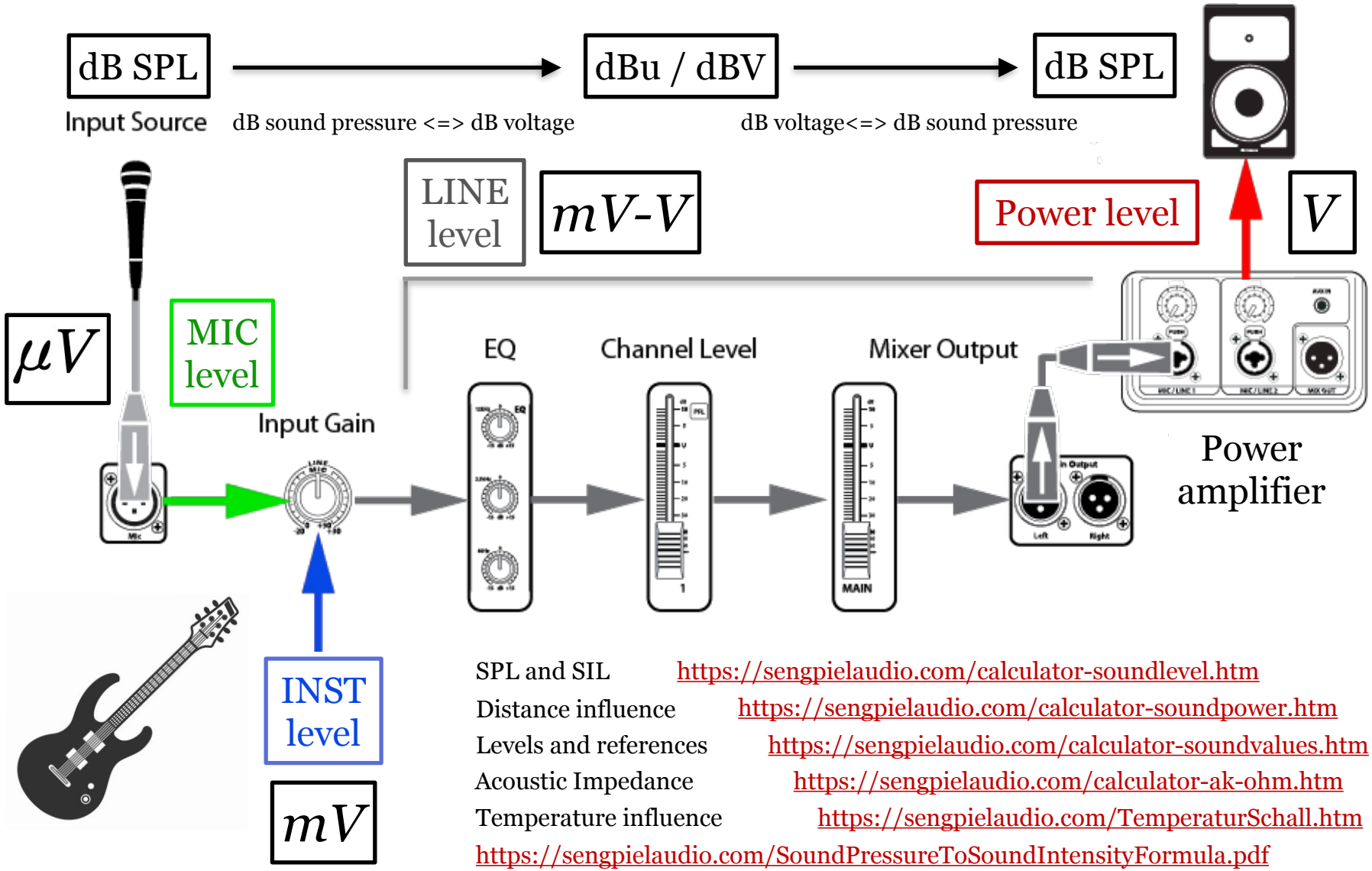
- Comb filtering - delay <10 - 50ms (depending on source and relative levels)
- Hass effect - Threshold of eco ~ delay > 50ms (non-impulsive sources)
- Franssen effect - Our ear lie!
- Cocktail party effect - Spatial and spectral filtering allow intelligibility in noisy environments

*figure 417: Utilising the Haas effect to provide localisation with large PA's*




BALLOU, Glen. *Handbook for sound engineers.* Taylor & Francis, 2013, p60.

# Electro-acoustic duality




SPL and SIL https://sengpielaudio.com/calculator-soundlevel.htm
Distance influence https://sengpielaudio.com/calculator-soundpower.htm
Levels and references https://sengpielaudio.com/calculator-soundvalues.htm
Acoustic Impedance https://sengpielaudio.com/calculator-ak-ohm.htm
Temperature influence https://sengpielaudio.com/TemperaturSchall.htm
https://sengpielaudio.com/SoundPressureToSoundIntensityFormula.pdf

# Decibels

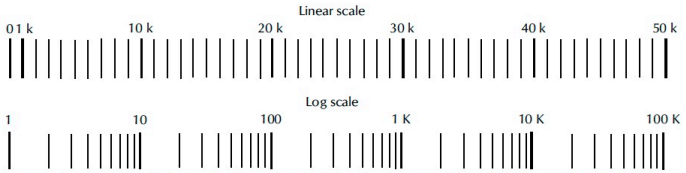



**Figure 2-1.** A logarithmic scale has its increments marked by a fixed ratio, in this case 10 to 1, forming a more compact representation than a linear scale. Courtesy Syn-Aud-Con.

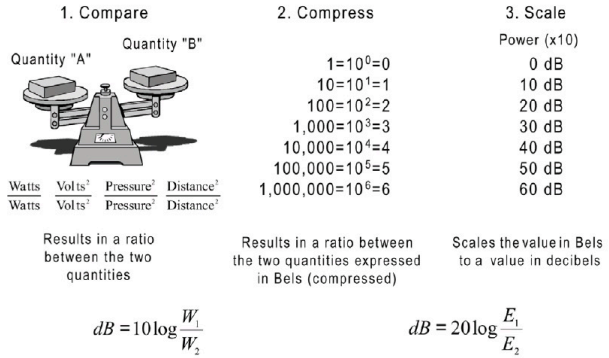



**Figure 2-2.** The steps to performing a decibel conversion are outlined. Courtesy Syn-Aud-Con.

BALLOU, Glen.
*Handbook for sound engineers.*
Taylor & Francis, 2013, p24.

RUMSEY, Francis; MCCORMICK, Tim.
*Sound and recording: an introduction*.
CRC Press, 2012, p17.

## FACT FILE 1.2 THE DECIBEL

### Basic Decibels

The decibel is based on the logarithm of the ratio between two numbers. It describes how much larger or smaller one value is than the other. It can also be used as an absolute unit of measurement if the reference value is fixed and known. Some standardized references have been established for decibel scales in different fields of sound engineering (see below).

The decibel is strictly ten times the logarithm to the base ten of the ratio between the powers of two signals:

$$dB = 10 \log_{10}(P_1/P_2)$$

For example, the difference in decibels between a signal with a power of 1 watt and one of 2 watts is $10\log(2/1) = 3$dB.

If the decibel is used to compare values other than signal powers, the relationship to signal power must be taken into account. Voltage has a square relationship to power (from Ohm's law: $W = V^2/R$); thus to compare two voltages:

$$dB = 10 \log(V_1^2/V_2^2), \text{ or } 10 \log(V_1/V_2)^2, \text{ or } 20 \log(V_1/V_2)$$

For example, the difference in decibels between a signal with a voltage of 1 volt and one of 2 volts is $20\log(2/1) = 6$dB. So a doubling in voltage gives rise to an increase of 6dB, and a doubling in power gives rise to an increase of 3dB. A similar relationship applies to acoustical sound pressure (analogous to electrical voltage) and sound power (analogous to electrical power).

### Decibels with a Reference

If a signal level is quoted in decibels, then a reference must normally be given, otherwise the figure means nothing; e.g. 'Signal level = 47dB' cannot have a meaning unless one knows that the signal is 47dB above a known point. ' +8dB ref. 1 volt' has a meaning since one now knows that the level is 8dB higher than 1 volt, and thus one could calculate the voltage of the signal.

There are exceptions in practice, since in some fields a reference level is accepted as implicit. Sound pressure levels (SPLs) are an example, since the reference level is defined worldwide as $2 \times 10^{-5}$N m$^{-2}$ (20 μPa). Thus to state 'SPL = 77 dB' is probably acceptable, although confusion can still arise due to misunderstandings over such things as weighting curves (see Fact File 1.4). In sound recording, 0dB or 'zero level' is a nominal reference level used for aligning equipment and setting recording levels, often corresponding to 0.775 volts (0dBu) although this is subject to variations in studio centers in different locations. (Some studios use 4dBu as their reference level, for example.) '0dB' does not mean 'no signal', it means that the signal concerned is at the same level as the reference.

Often a letter is placed after 'dB' to denote the reference standard in use (e.g. 'dBm'), and a number of standard abbreviations are in use, some examples of which are given below. Sometimes the suffix denotes a particular frequency weighting characteristic used in the measurement of noise (e.g. 'dBA').

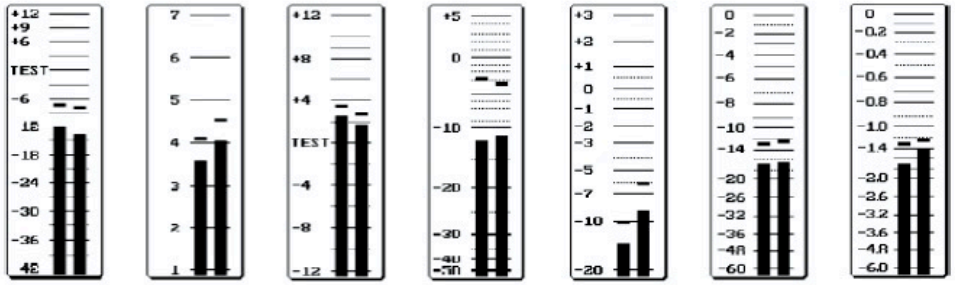
| Abbrev. | Ref. level |
|---|---|
| dBV | 1 volt |
| dBu | 0.775 volt (Europe) |
| dBv | 0.775 volt (USA) |
| dBm | 1 milliwatt (see Chapter 12) |
| dBA | dB SPL, A-weighted response |

### Useful Decibel Ratios to Remember (Voltages or SPLs)

It is more common to deal in terms of voltage or SPL ratios than power ratios in audio systems. Here are some useful dB equivalents of different voltage or SPL relationships and multiplication factors:

| dB | Multiplication factor |
|---|---|
| 0 | 1 |
| +3 | √2 |
| +6 | 2 |
| +20 | 10 |
| +60 | 1000 |

# Metering standards




- time weighting
  - PPM standard (5-10ms)
  - VU standard (300ms)
  - *at a VU meter peaks* will rest -6dB to -12dB below!

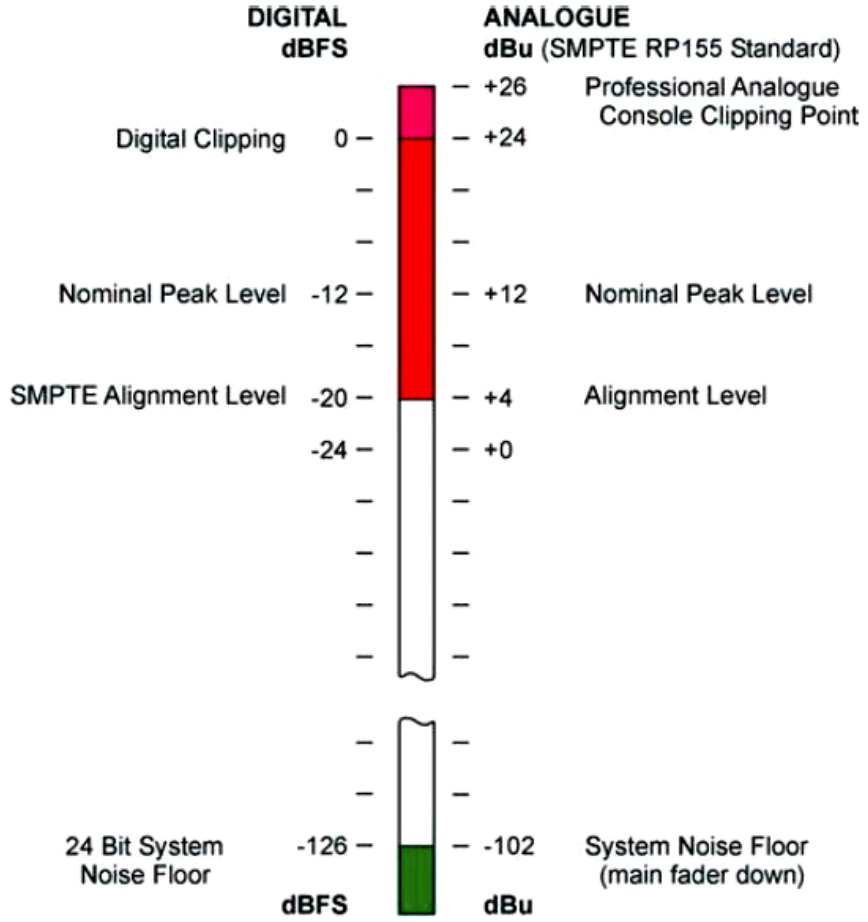
| escale | reference | usage/notes |
|---|---|---|
| dBµW | 1µW | |
| dBm | 1mW at 600Ω | $\sqrt{(1mW*600\Omega)} \approx 0{,}775V$ |
| dBu | 0,775V RMS | former dBv |
| dBV | 1V | |
| dBW | 1W | |
| dBFS | digital full scale | negative scale |

| volts | dBu | IEC I Nordic | IEC IIa BBC | IEC IIb | DIN | VU Standard | VU North Am., Australia | VU France | EBU AD/DA |
|---|---|---|---|---|---|---|---|---|---|
| 12.28 | 24 | | | | | | | | |
| 10.95 | 23 | | | | | | | | |
| 9.76 | 22 | | | | | | | | |
| 8.70 | 21 | | | | | | | | |
| 7.75 | 20 | | | | | | | | |
| 6.91 | 19 | | | | | | | | |
| 6.16 | 18 | | | | | | | | |
| 5.49 | 17 | | | | | | | | |
| 4.89 | 16 | | | | | | | | |
| 4.36 | 15 | | | | | | | | |
| 3.88 | 14 | | | | | | | | |
| 3.46 | 13 | | | | | | | | |
| 3.09 | 12 | 12 | 7 | 12 | 5 | | | | 0 |
| 2.75 | 11 | 11 | | 11 | | | | | -1 |
| 2.45 | 10 | 10 | | 10 | 4 | | | | -2 |
| 2.18 | 9 | 9 | | 9 | 3 | | | | -3 |
| 1.95 | 8 | 8 | 6 | 8 | 2 | | | | -4 |
| 1.74 | 7 | 7 | | 7 | 1 | +3 | | | -5 |
| 1.55 | 6 | 6 | | 6 | 0 | +2 | | | -10 |
| 1.38 | 5 | 5 | | 5 | -1 | +1 | | | -11 |
| 1.23 | 4 | 4 | 5 | 4 | -2 | 0 | | | -12 |
| 1.10 | 3 | 3 | | 3 | -3 | -1 | +3 | | -13 |
| 0.976 | 2 | 2 | | 2 | -4 | -2 | +2 | | -14 |
| 0.870 | 1 | 1 | | 1 | -5 | -3 | +1 | +3 | -15 |
| 0.775 | 0 | test | 4 | test | -6 | -4 | 0 | +2 | -16 |
| 0.691 | -1 | -1 | | -1 | -7 | -5 | -1 | +1 | -17 |
| 0.616 | -2 | -2 | | -2 | -8 | -6 | -2 | 0 | -18 |
| 0.549 | -3 | -3 | | -3 | test | -7 | -3 | -1 | -19 |
| 0.489 | -4 | -4 | 3 | -4 | -10 | -8 | -4 | -2 | -20 |
| 0.436 | -5 | -5 | | -5 | -11 | -9 | -5 | -3 | -21 |
| 0.388 | -6 | -6 | | -6 | -12 | -10 | -6 | -4 | -22 |
| 0.346 | -7 | -7 | | -7 | -13 | -11 | -7 | -5 | -23 |
| 0.309 | -8 | -8 | 2 | -8 | -14 | -12 | -8 | -6 | -24 |
| 0.275 | -9 | -9 | | -9 | -15 | -13 | -9 | -7 | -25 |
| 0.245 | -10 | -10 | | -10 | -16 | -14 | -10 | -8 | -26 |
| 0.218 | -11 | -11 | | -11 | -17 | -15 | -11 | -9 | -27 |
| 0.195 | -12 | -12 | 1 | -12 | -18 | -16 | -12 | -10 | -28 |
| 0.174 | -13 | -13 | | | -19 | -17 | -13 | -11 | -29 |
| 0.155 | -14 | -14 | | | -20 | -18 | -14 | -12 | -30 |
| 0.138 | -15 | -15 | | | -21 | -19 | -15 | -13 | -31 |
| 0.123 | -16 | -16 | | | -22 | -20 | -16 | -14 | -32 |
| 0.109 | -17 | -17 | | | -23 | | -17 | -15 | -33 |
| 98 m | -18 | -18 | | | -24 | | -18 | -16 | -34 |
| 87 m | -19 | -19 | | | -25 | | -19 | -17 | -35 |
| 78 m | -20 | -20 | | | -26 | | -20 | -18 | -36 |
| 69 m | -21 | -21 | | | -27 | | | -19 | -37 |
| 61 m | -22 | -22 | | | -28 | | | -20 | -38 |
| 55 m | -23 | -23 | | | -29 | | | | -39 |
| 49 m | -24 | -24 | | | -30 | | | | -40 |
| 44 m | -25 | -25 | | | -31 | | | | -41 |
| 39 m | -26 | -26 | | | -32 | | | | -42 |
| 35 m | -27 | -27 | | | -33 | | | | -43 |
| 31 m | -28 | -28 | | | -34 | | | | -44 |
| 28 m | -29 | -29 | | | -35 | | | | -45 |
| ⇓ | ⇓ | ⇓ | ⇓ | ⇓ | ⇓ | | | | -46 |
| - | - | -42 | -∞ | -∞ | -50 | | | | -47 |
| | | | | | | | | | ⇓ |
| | | | | | | | | | - |

# EBU R68 and SMPTE RP155

*Relationship between analog signal level and digital coding in converters as per the EBU and SMPTE standards respectively.*

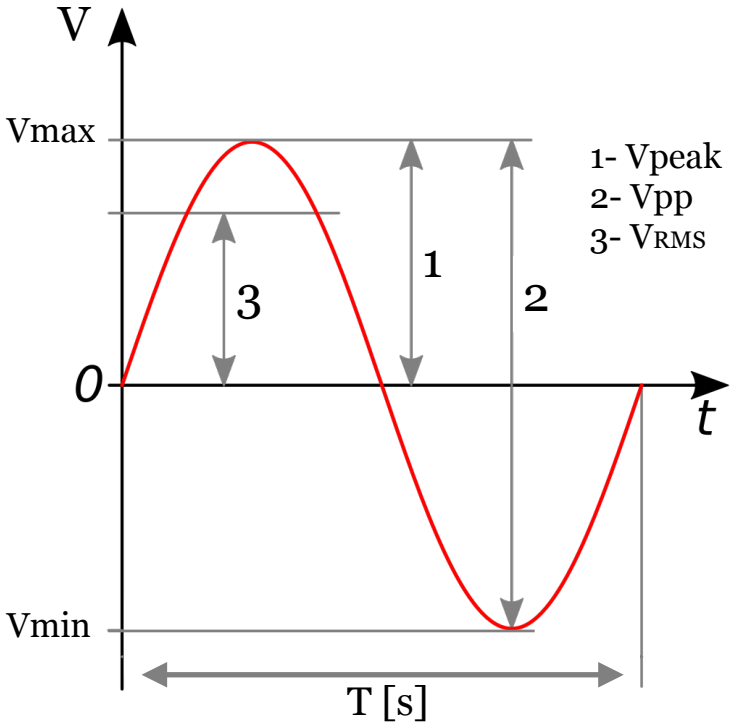
| | EBU R68 | | | SMPTE RP155 | | | |
|---|---|---|---|---|---|---|---|
| | dBFS | dBu | Volts | dBFS | dBu | Volts | |
| | 0 | 18 | 6.16 | 0 | 24 | 12.28 | Max level |
| | -1 | 17 | 5.49 | -1 | 23 | 10.95 | |
| | -2 | 16 | 4.89 | -2 | 22 | 9.76 | |
| | -3 | 15 | 4.36 | -3 | 21 | 8.70 | |
| | -4 | 14 | 3.88 | -4 | 20 | 7.75 | |
| | -5 | 13 | 3.46 | -5 | 19 | 6.91 | |
| | -6 | 12 | 3.09 | -6 | 18 | 6.16 | |
| | -7 | 11 | 2.75 | -7 | 17 | 5.49 | |
| | -8 | 10 | 2.45 | -8 | 16 | 4.89 | |
| | -9 | 9 | 2.18 | -9 | 15 | 4.36 | |
| | -10 | 8 | 1.95 | -10 | 14 | 3.88 | |
| | -11 | 7 | 1.74 | -11 | 13 | 3.46 | |
| | -12 | 6 | 1.55 | -12 | 12 | 3.09 | |
| | -13 | 5 | 1.38 | -13 | 11 | 2.75 | |
| | -14 | 4 | 1.23 | -14 | 10 | 2.45 | |
| | -15 | 3 | 1.09 | -15 | 9 | 2.18 | |
| | -16 | 2 | 0.976 | -16 | 8 | 1.95 | |
| | -17 | 1 | 0.870 | -17 | 7 | 1.74 | |
| Test | -18 | 0 | 0.775 | -18 | 6 | 1.55 | |
| | -19 | -1 | 0.691 | -19 | 5 | 1.38 | |
| | -20 | -2 | 0.616 | -20 | 4 | 1.23 | 0 VU |
| | -21 | -3 | 0.549 | -21 | 3 | 1.09 | |
| | -22 | -4 | 0.489 | -22 | 2 | 0.976 | |
| | -23 | -5 | 0.436 | -23 | 1 | 0.870 | |
| | -24 | -6 | 0.388 | -24 | 0 | 0.775 | |
| | -25 | -7 | 0.346 | -25 | -1 | 0.691 | |
| | -26 | -8 | 0.309 | -26 | -2 | 0.616 | |
| | -27 | -9 | 0.275 | -27 | -3 | 0.549 | |
| | -28 | -10 | 0.245 | -28 | -4 | 0.489 | |
| | -29 | -11 | 0.218 | -29 | -5 | 0.436 | |
| | -30 | -12 | 0.195 | -30 | -6 | 0.388 | |
| | -31 | -13 | 0.174 | -31 | -7 | 0.346 | |
| | -32 | -14 | 0.155 | -32 | -8 | 0.309 | |
| | -33 | -15 | 0.138 | -33 | -9 | 0.275 | |
| | -34 | -16 | 0.123 | -34 | -10 | 0.245 | |
| | -35 | -17 | 0.109 | -35 | -11 | 0.218 | |
| | -36 | -18 | 0.098 | -36 | -12 | 0.195 | |
| | -37 | -19 | 0.087 | -37 | -13 | 0.174 | |
| | -38 | -20 | 0.078 | -38 | -14 | 0.155 | |
| | -39 | -21 | 0.069 | -39 | -15 | 0.138 | |
| | -40 | -22 | 0.062 | -40 | -16 | 0.123 | |
| | -41 | -23 | 0.055 | -41 | -17 | 0.109 | |
| | -42 | -24 | 0.049 | -42 | -18 | 0.098 | |

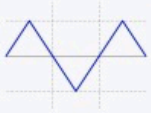
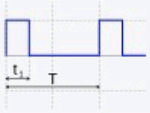



A comparison between traditional professional analogue console signal levels and the SMPTE recommended digital equivalents (Rp155).

# Peak and RMS levels. Crest factor

For sinusiods: $V_{RMS} = \frac{V_{peak}}{\sqrt{2}}$



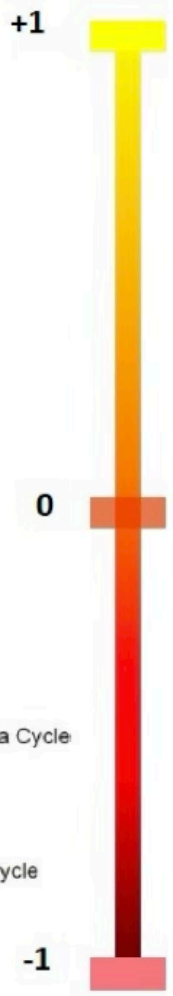

| Wave type | Waveform | RMS value | Crest factor | PAPR (dB) |
|---|---|---|---|---|
| DC | | 1 | 1 | 0.0 dB |
| Sine wave | | $\frac{1}{\sqrt{2}} \approx 0.707$[5] | $\sqrt{2} \approx 1.414$ | 3.01 dB |
| Full-wave rectified sine | | $\frac{1}{\sqrt{2}} \approx 0.707$[5] | $\sqrt{2} \approx 1.414$ | 3.01 dB |
| Half-wave rectified sine | | $\frac{1}{2} = 0.5$[5] | 2 | 6.02 dB |
| Triangle wave |  | $\frac{1}{\sqrt{3}} \approx 0.577$ | $\sqrt{3} \approx 1.732$ | 4.77 dB |
| Square wave | | 1 | 1 | 0 dB |
| PWM signal<br>V(t) ≥ 0.0 V |  | $\sqrt{\frac{t_1}{T}}$[5] | $\sqrt{\frac{T}{t_1}}$ | $20 \log \left(\frac{T}{t_1}\right)$ dB |

General expression: $Crest\ factor = \frac{Vp}{V_{RMS}}$

# Phase relationship and coherence

- phase relationship = $\cos(\theta_1-\theta_2)$

  - $\cos(\theta_1-\theta_2) = +1 \rightarrow (\theta_1-\theta_2) = 0°$ → phase correlation → **Coherent**
  - $\cos(\theta_1-\theta_2) = 0 \rightarrow (\theta_1-\theta_2) = 90°$ → quadrature → **Non-coherent**
  - $\cos(\theta_1-\theta_2) = -1 \rightarrow (\theta_1-\theta_2) = 180°$ → phase correlation → **Coherent**



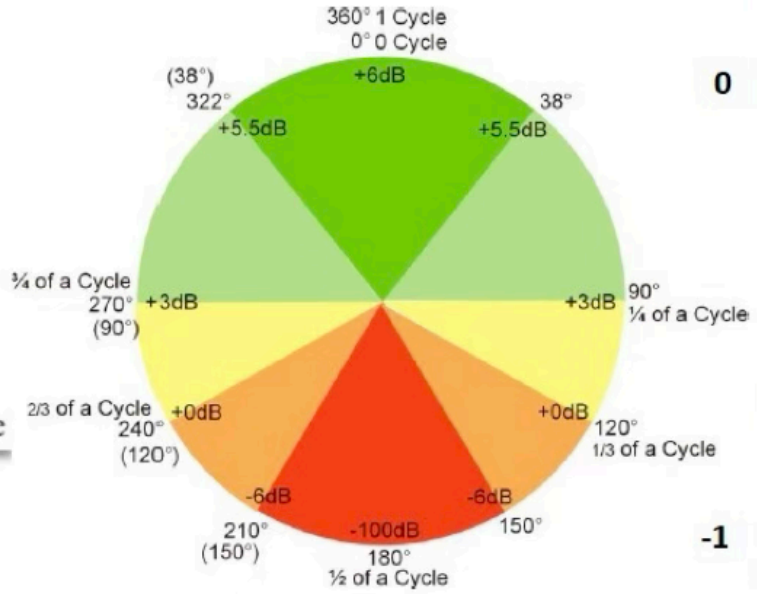

  - slow integration time ≈ 600ms
  - average level vs. RMS level

$$A_{Average} = \frac{1}{T}\int_0^T |a|\,dt$$

where:
$T$ = Period
$a$ = Amplitude

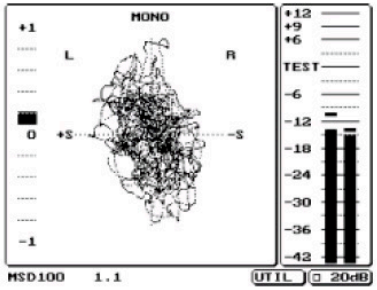
$$A_{RMS} = \sqrt{\frac{1}{T}\int_0^T a^2(t)\,dt}$$

where:
$T$ = Period
$a$ = Amplitude as a function of time

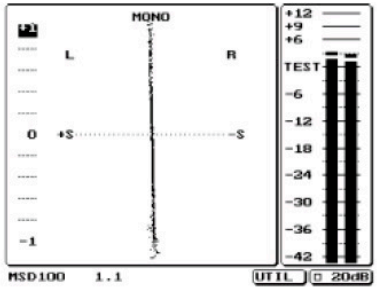



  - crest factor = peak level / RMS level

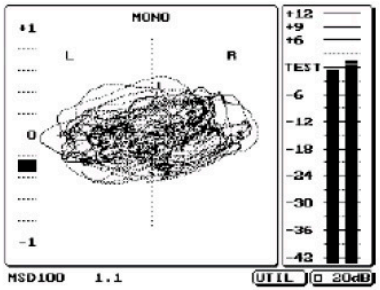
# Goniometer/Vectorscope



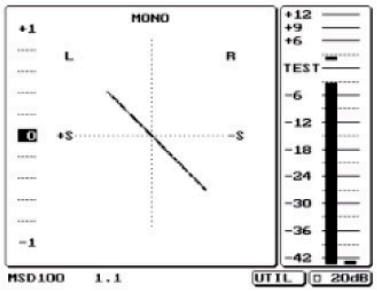
Stereo Signal



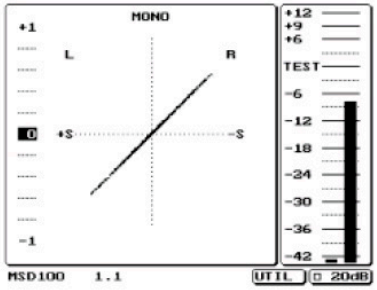
Mono Signal



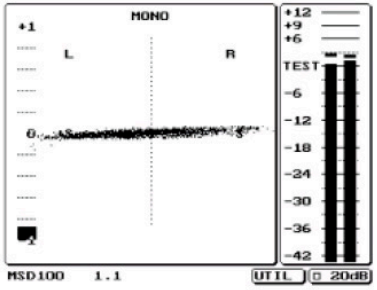
Reversed Phase Stereo Signal



Left Signal



Right Signal



Reversed Phase Mono Signal

## Lissajous curves

$$x = A\sin(at + \delta)$$

$$y = B\sin(bt)$$

a/b = number of lobes ->Ellipse for a/b = 1

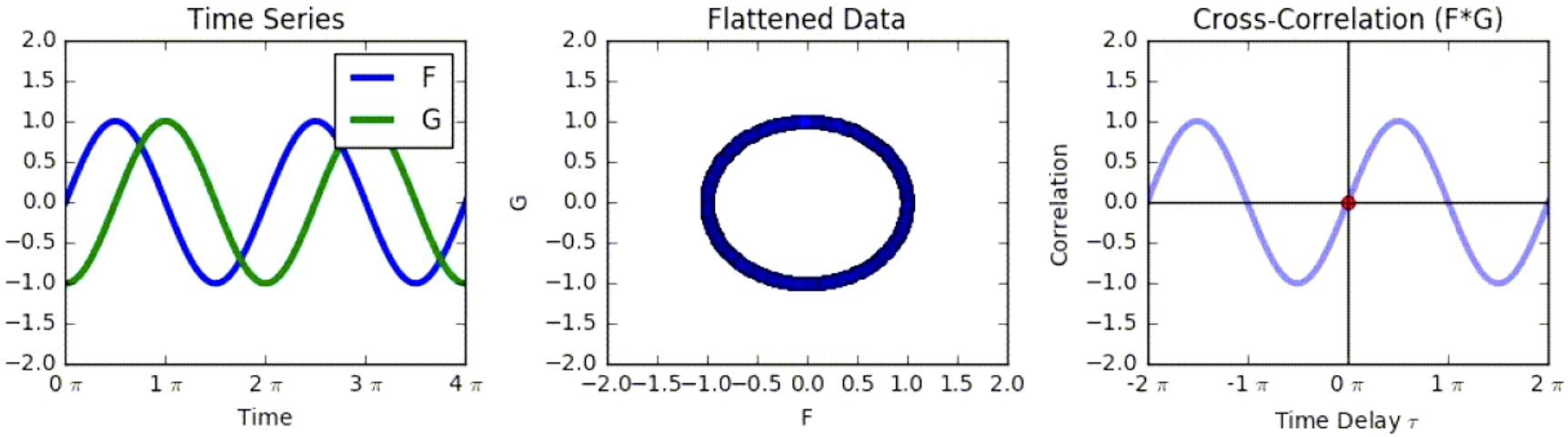
https://es.wikipedia.org/wiki/Curva_de_Lissajous

# Cross-correlation

Video: Measuring Time Delay using Cross Correlation

Cross-correlation $$R(\tau) = \int_{-\infty}^{+\infty} x(t)\, y(t+\tau)\, dt$$

Auto cross-correlation when x(t) = y(t)




By Divergentdata - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=57768455

# Spectrum analysis. FFT

## Fast Fourier Transform

- 1024-band FFT (44Hz-20KHz)
  - $f_{resoution} = f_s/N$
  - at $f_s$=44100Hz, N=1024→$f_{resoution}$= 43Hz

fs = Sample rate
N = number of bands

- 1/3 octave (20Hz-16KHz)

| ISO Band Numbers | Octave Band Center Frequency | One-Third Octave Band Center Frequencies |
|---|---|---|
| 11, 12, 13 | 16 Hz | 12.5 Hz, 16 Hz, 20 Hz |
| 14, 15, 16 | 31.5 Hz | 25 Hz, 31 Hz, 40 Hz |
| 17, 18, 19 | 63 Hz | 50 Hz, 63 Hz, 80 Hz |
| 20, 21, 22 | 125 Hz | 100 Hz, 125 Hz, 160 Hz |
| 23, 24, 25 | 250 Hz | 200 Hz, 250 Hz, 315 Hz |
| 26, 27, 28 | 500 Hz | 400 Hz, 500 Hz, 630 Hz |
| 29, 30, 31 | 1000 Hz | 800 Hz, 1000 Hz, 1250 Hz |
| 32, 33, 34 | 2000 Hz | 1600 Hz, 2000 Hz, 2500 Hz |
| 35, 36, 37 | 4000 Hz | 3150 Hz, 4000 Hz, 5000 Hz |
| 38, 39, 40 | 8000 Hz | 6300 Hz, 8000 Hz, 10000 Hz |
| 41, 42, 43 | 16000 Hz | 12500 Hz, 16000 Hz, 20000 Hz |

REW RTA - Pink noise measurement.
Roland DS-7. SPL vs frequency

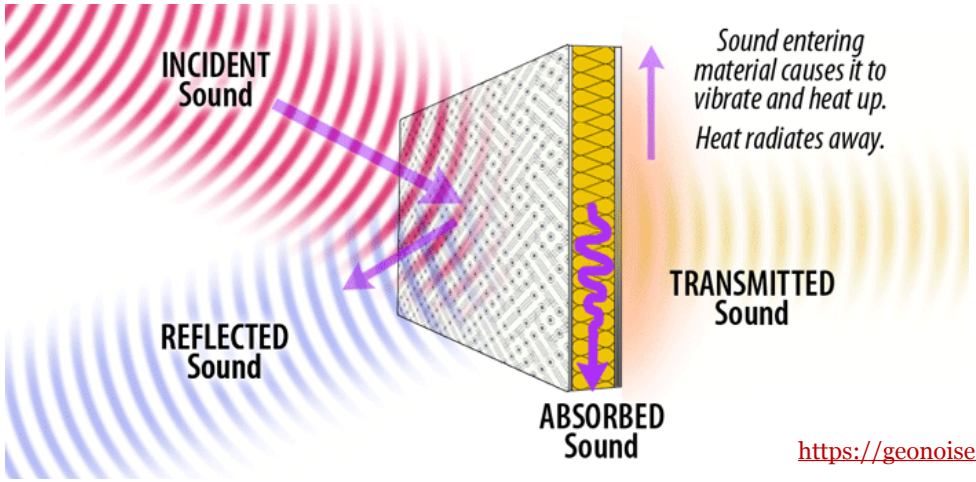
https://sengpielaudio.com/calculator-octave.htm

# Magnitude and phase analysis

REW - Logarithmic sweep measurement. Roland DS-7. SPL and Phase vs frequency.

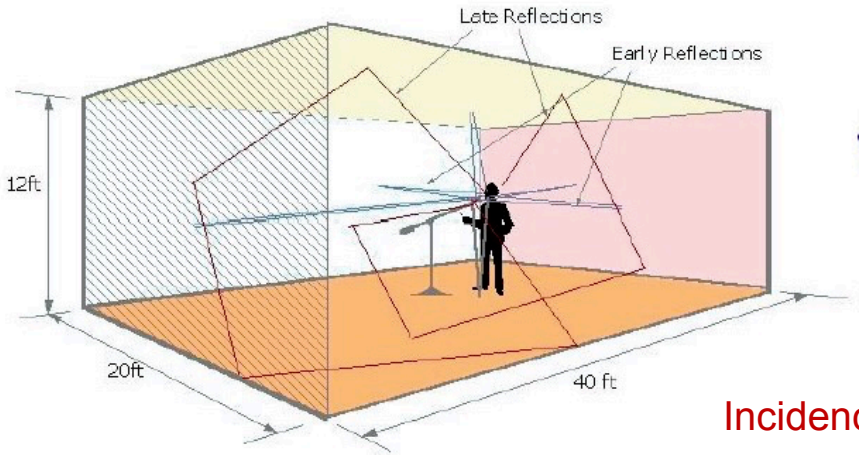
# Absorption, reflection, refraction, and diffraction

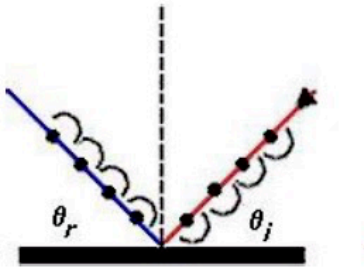
- **absorption:** energy dissipation due to transduction, from kinetic to calorific energy




https://geonoise.asia/sound-absorption/

- **reflection**: changes in the direction of propagation due to physical obstacles




**Snell law 1:**
Incidence angle = reflected angle

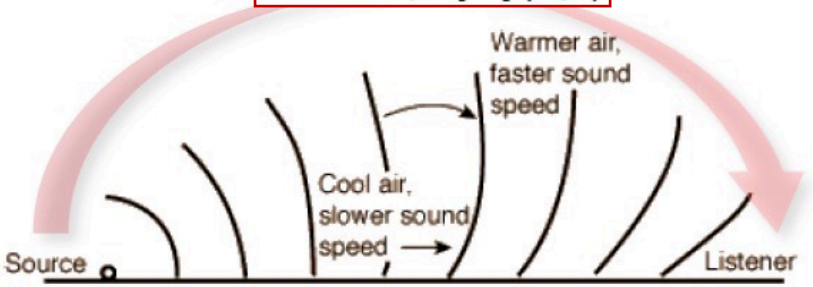
# Absorption, reflection, refraction, and diffraction

- **refraction**: changes in propagation speed due to
  - changes in medium density (kg/m$^3$)
  - temperature changes

$$\text{sound speed (m/s)} = \sqrt{\frac{\text{stiffness (Young's modulus, N/m2)}}{\text{mass density (Kg/m3)}}}$$

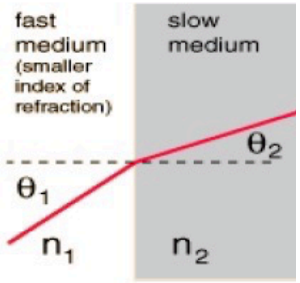
greater medium density *and/or* lower temperature increase $n_2$ → approach propagation direction to the boundary *normal*

$v \approx 331+0{,}6T[ºC]$ (m/s)

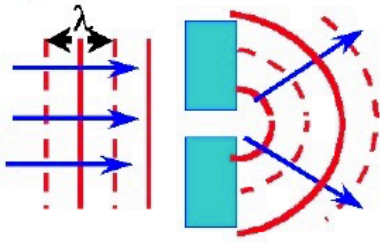


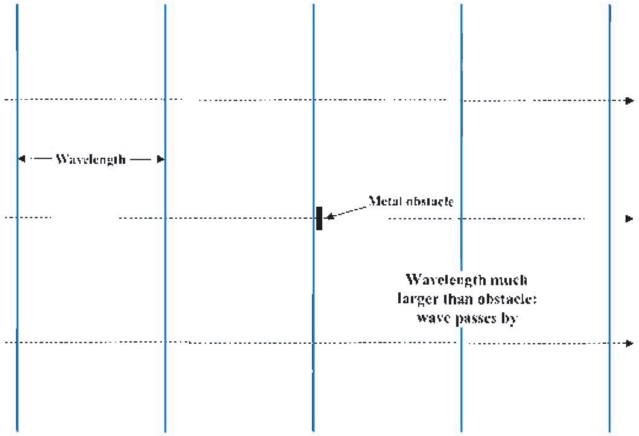

**Snell law 2:**

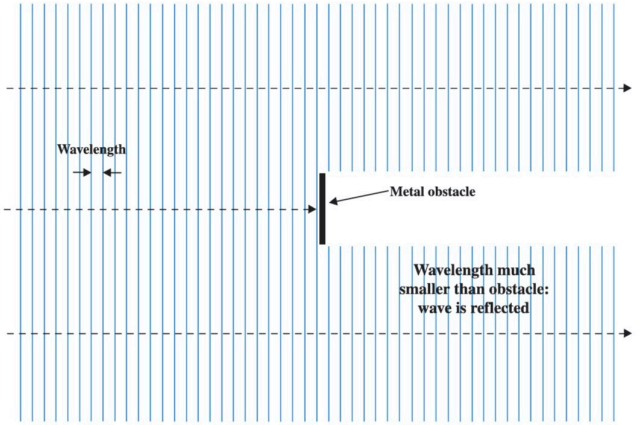
$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$




- **diffraction**: spreading of waves as they pass through or around an obstacle



# Reflection vs Diffusion

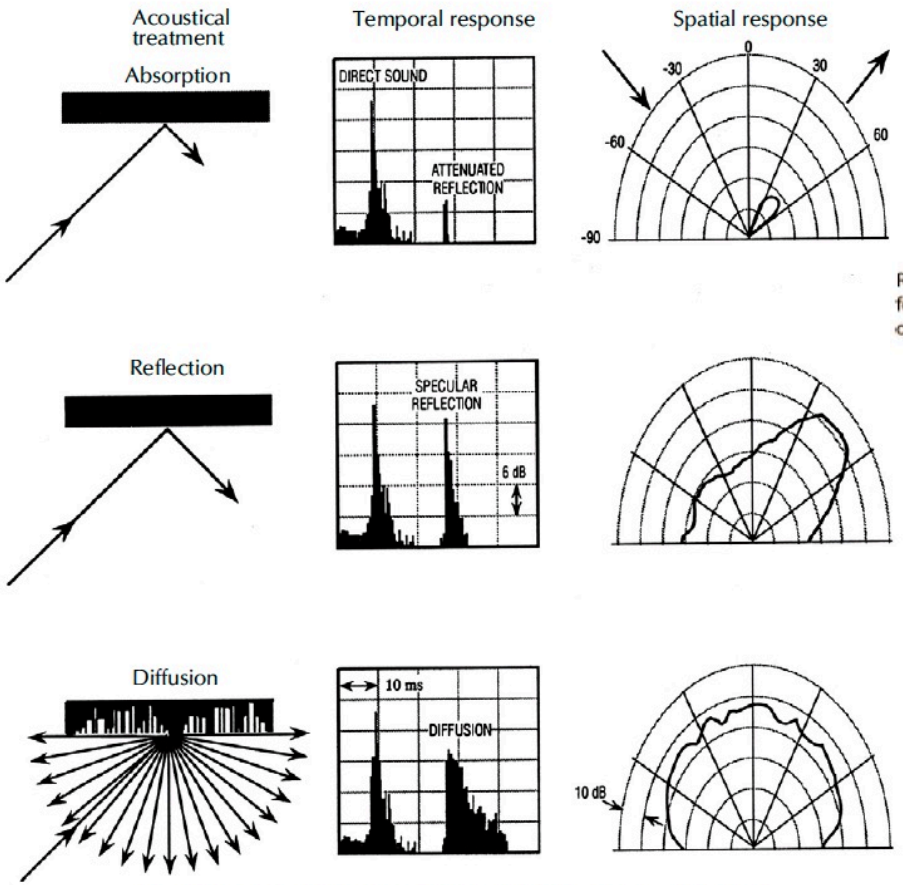
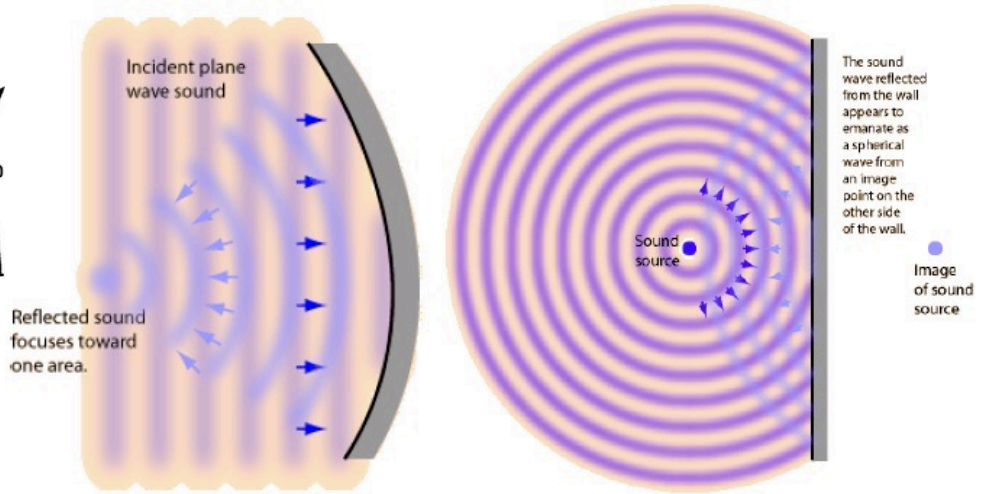



**Figure 5-1.** Comparison of absorption reflection and diffusion.



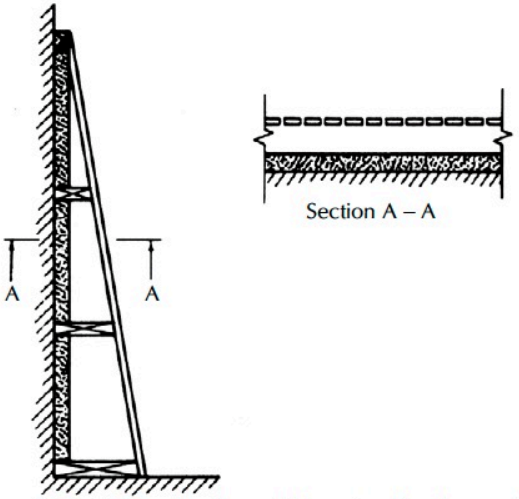



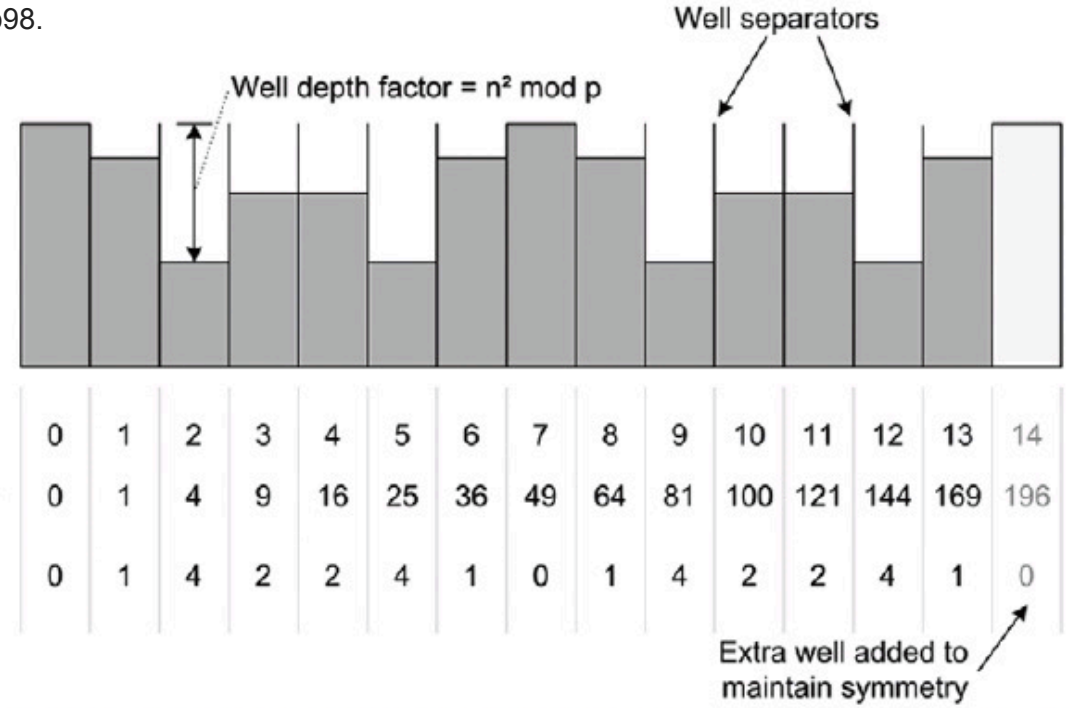

**Figure 5-28.** A slat absorber with varying depth to widen the effective frequency range of absorption.

BALLOU, Glen. *Handbook for sound engineers.* Taylor & Francis, 2013, pp 98,113.

# Quadratic Residue Diffusers (QRD)

The depth of the wells is calculated from a quadratic residue sequence to scatter the sound waves in different directions. This process spreads out the sound energy spatially and temporally within the room, reducing slap echoes and comb filtering associated with strong reflections.

BALLOU, Glen.
*Handbook for sound engineers.*
Taylor & Francis, 2013, p98.

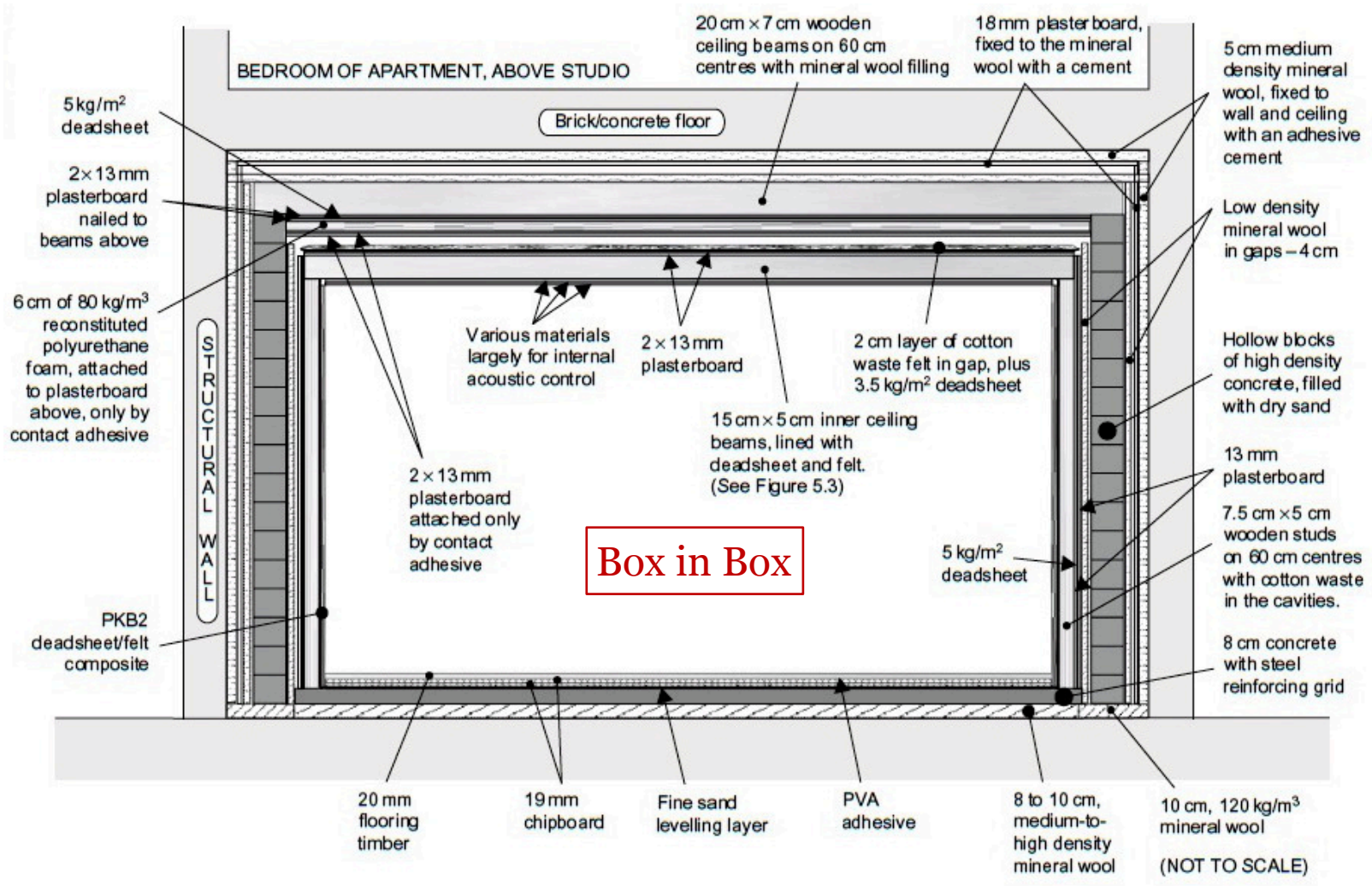



| n | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| n² | 0 | 1 | 4 | 9 | 16 | 25 | 36 | 49 | 64 | 81 | 100 | 121 | 144 | 169 | 196 |
| n² mod p (p = 7) | 0 | 1 | 4 | 2 | 2 | 4 | 1 | 0 | 1 | 4 | 2 | 2 | 4 | 1 | 0 |

**Figure 5-30.** A profile of two periods (with one extra well to maintain symmetry) of a QRD of prime number $p = 7$.

Example: n=3, 9 mod 7 = 2 (the reminder when 9 is divided by 7)
More about Quadratic residue sequences at: https://mathworld.wolfram.com/QuadraticResidue.html

# Acoustic Isolation. Soundproofing




SELF, Douglas. *Audio engineering explained*. Routledge, 2012, p584.

# Room modes

https://www.hunecke.de/en/calculators/room-eigenmodes.html

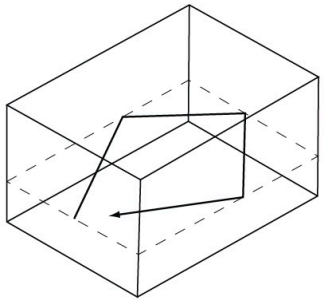


Figure 6-2. Tangential room modes.

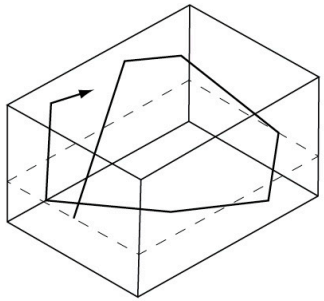
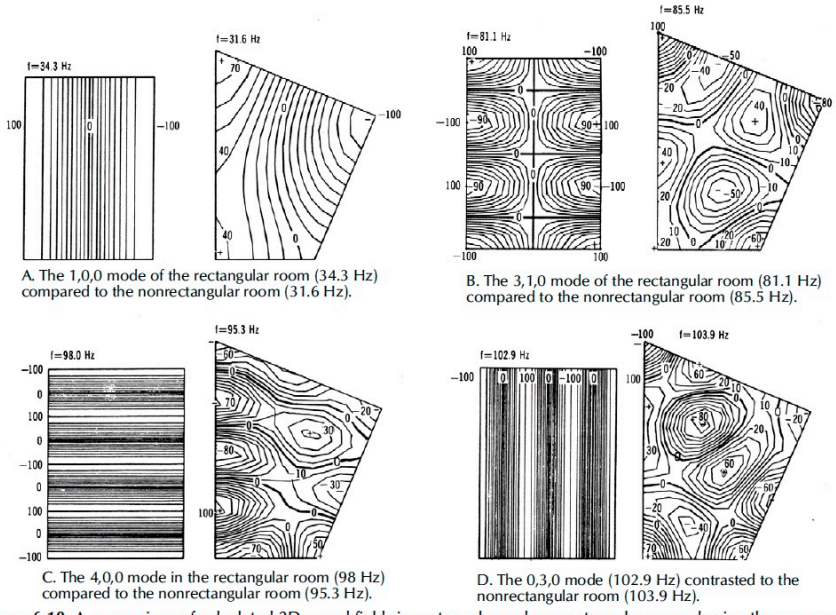


Figure 6-3. Oblique room modes.




A. The 1,0,0 mode of the rectangular room (34.3 Hz) compared to the nonrectangular room (31.6 Hz).

B. The 3,1,0 mode of the rectangular room (81.1 Hz) compared to the nonrectangular room (85.5 Hz).

C. The 4,0,0 mode in the rectangular room (98 Hz) compared to the nonrectangular room (95.3 Hz).

D. The 0,3,0 mode (102.9 Hz) contrasted to the nonrectangular room (103.9 Hz).

**Figure 6-10.** A comparison of calculated 2D sound fields in rectangular and nonrectangular rooms having the same areas. After Reference 8.

$$f = \frac{c}{2}\sqrt{\left(\frac{n_l}{l}\right)^2\left(\frac{n_w}{w}\right)^2\left(\frac{n_h}{h}\right)^2}$$

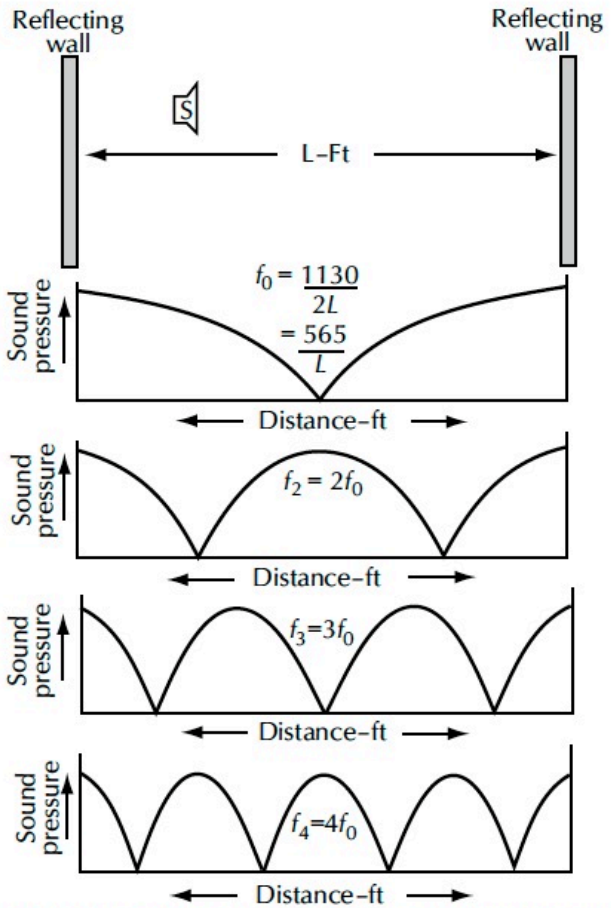



**Figure 6-1.** The simplest form of room resonance can be illustrated by two isolated, parallel, reflecting wall surfaces.

SELF, Douglas. *Audio engineering explained*. Routledge, 2012, p127, 128, 134.

# Acoustic conditioning. Reflection design

SELF, Douglas.
*Audio engineering explained.*
Routledge, 2012, p70.

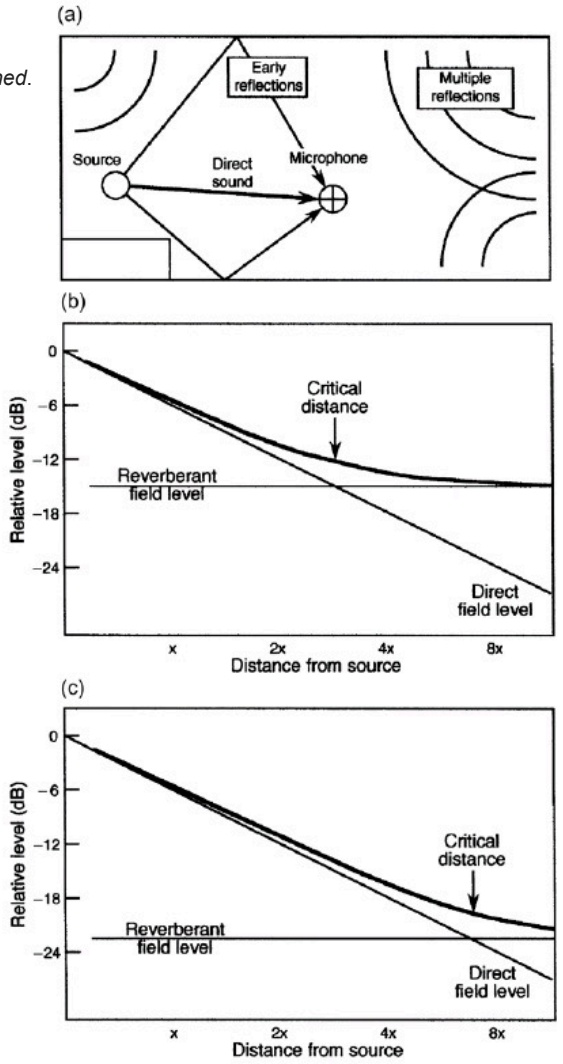
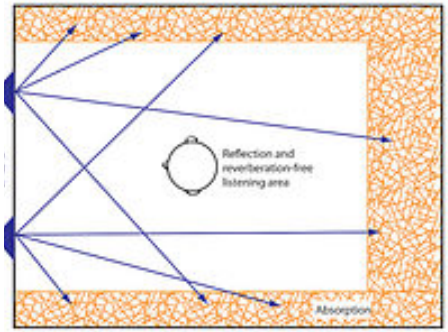
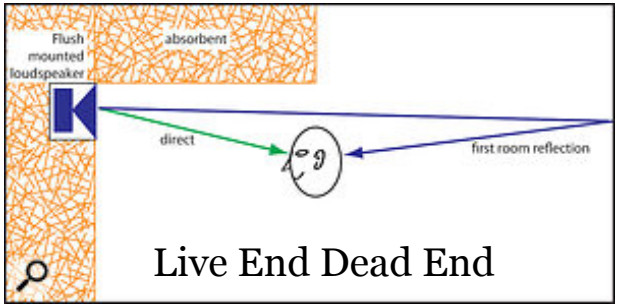



**FIGURE 3.8**
The reverberant field. Illustration of reflections in an enclosed space compared to direct sound at a variable distance from the sound source (a); interaction of direct and reverberant fields in a live space (b); interaction of direct and reverberant fields in a damped space (c).

https://www.soundonsound.com/techniques/sos-guide-control-room-design

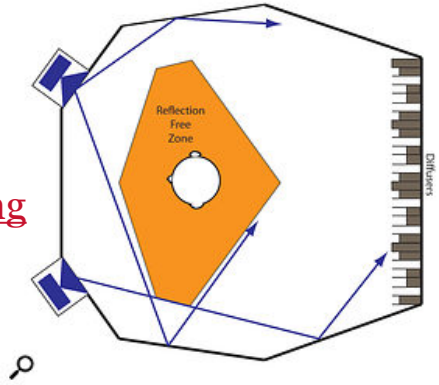
Non-environment



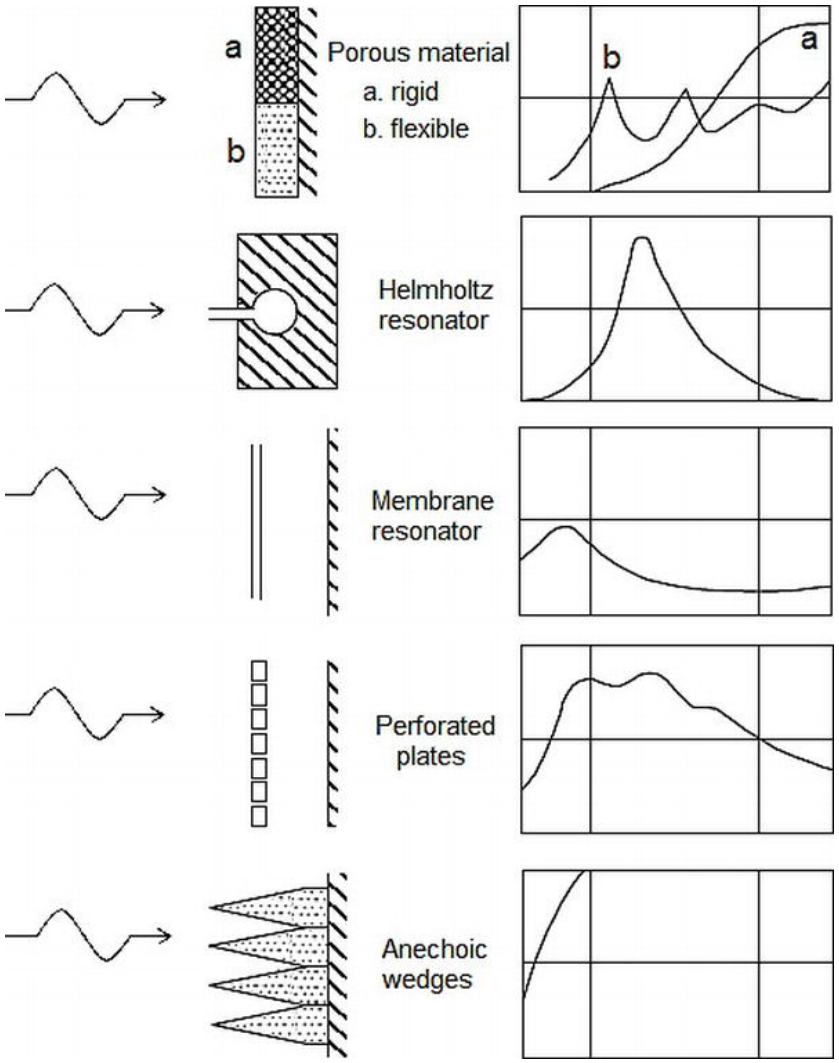

Reflection free zone

Early Sound Scattering



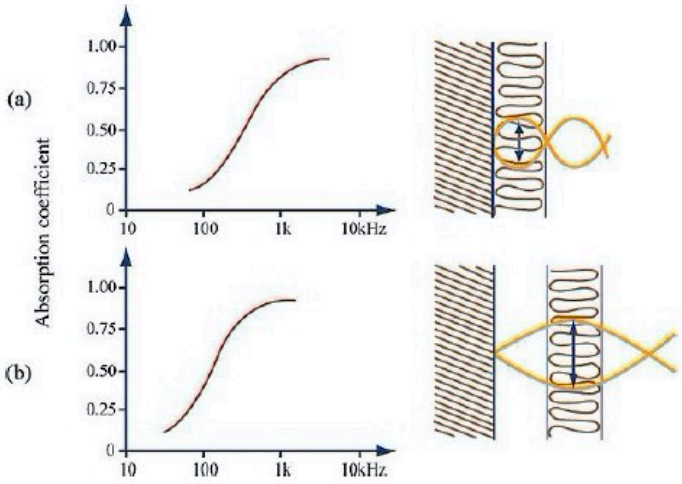
# Acoustic treatment

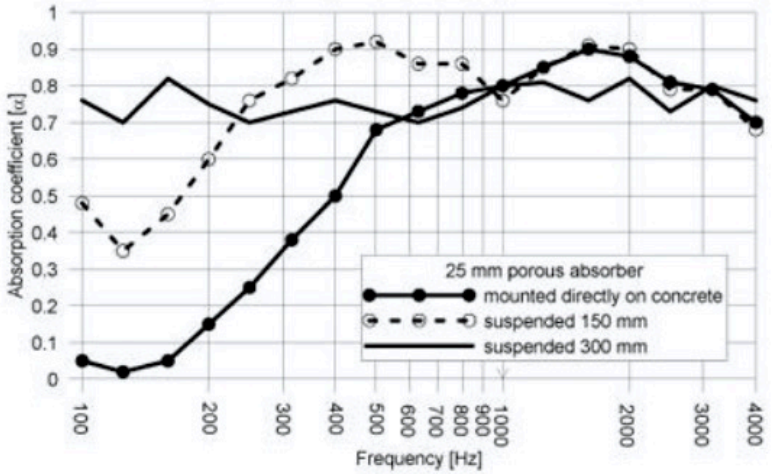
- previous issues to consider
  - isolation to external vibrations
  - dimensions and shape (room modes)
  - sweet spot location , sound propagation direction
  - different implementation approaches for LF (modal zone) and MF-HF
- different acoustic materials and solutions
  - reflectors
  - absorbers
  - diffusers, abfusers
- different design philosophies
  - LEDE (Live End – Death End)
  - RFZ (Reflection-Free Zone)
  - Non Environment (Philip Newell)
  - ESS (Early Sound Scattering)



# Acoustic materials

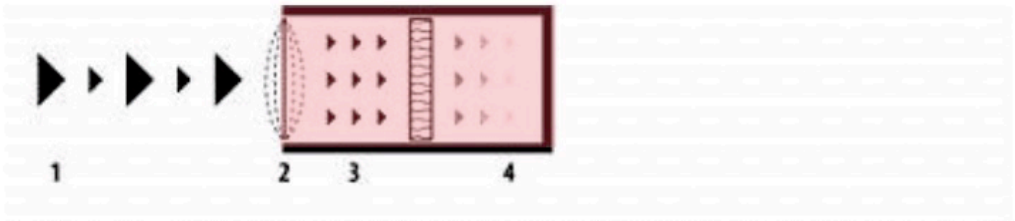
- absorption:
  - porous materials
    - fiberglass, rockwool, foam, cloth, etc.
  - kinetic energy is transduced into calorific energy
  - direct relationship among absortion bandwidth , wavelength and dimensions (width and depth) of acoustic materials
  - efectivity increases when moved away from reflective surfaces (walls)



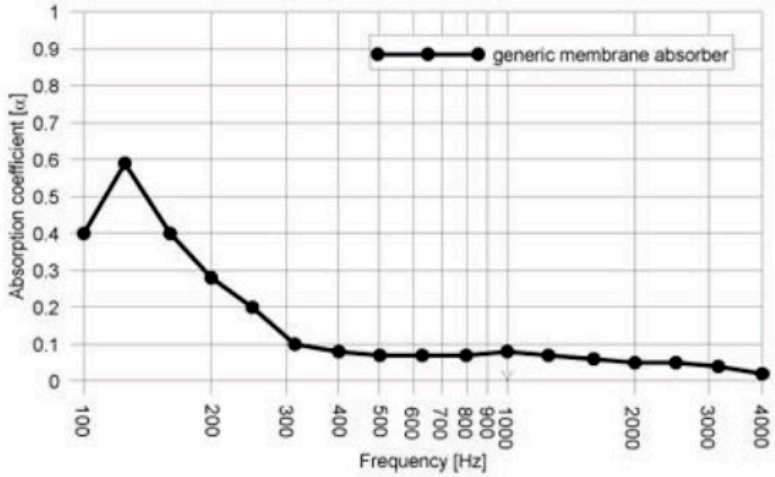



# Acoustic materials

- mebrane ressonators



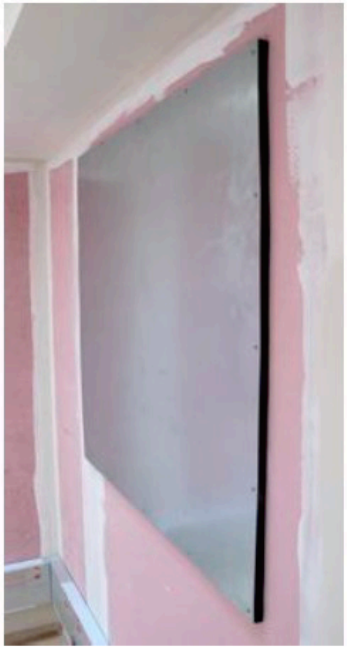
1-Sound incidence over the tuned membrane
2-Sympathy resonance
3-Sound traveling through the cavity
4-The absorber material attenuates the wavefront




$f_{ressonant} \approx 60/\sqrt{Md}$

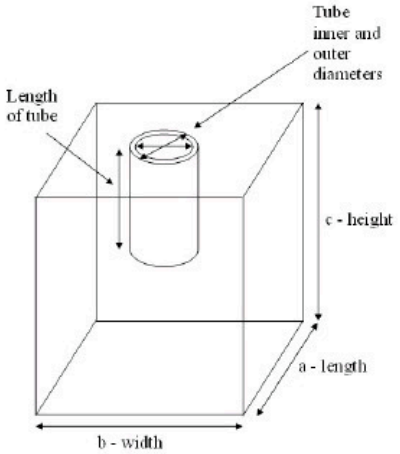
M = mass (Kg/m2),

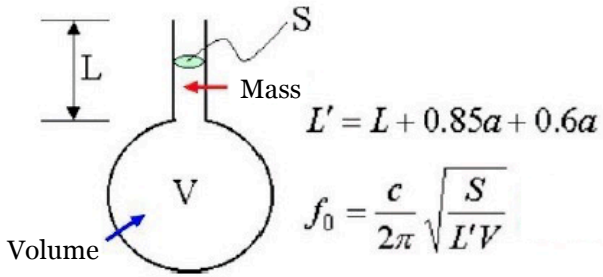
d = air chamber depth (m)



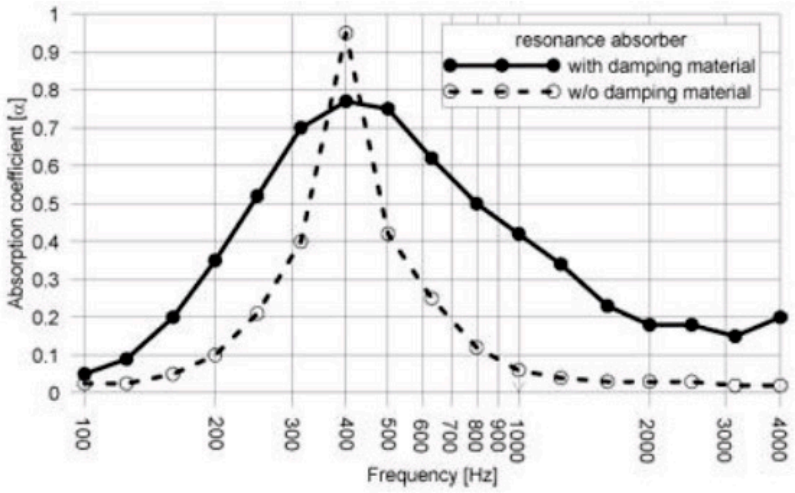
# Acoustic materials

- Helmholtz resonator
  - based on the mass-spring model
  - active method for perimetral low frequency cancelation at the ressonant frequency



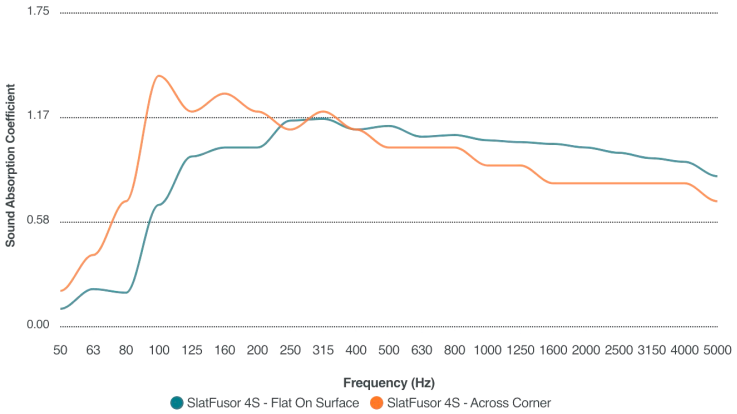

*fo* = Resonance frequency
c = speed of sound in gas
a = radius
S= Surface of the port
L= Longitude
L ' = Effective longitude
V = Volume



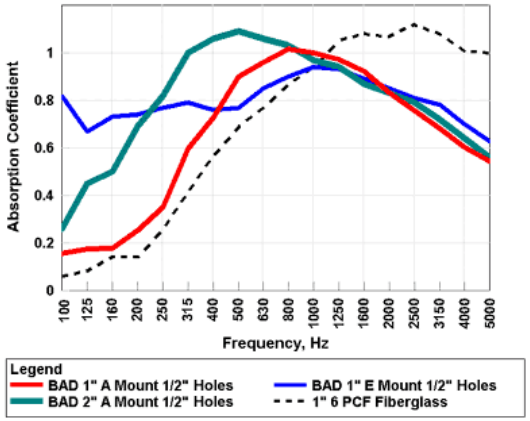
# Acoustic materials

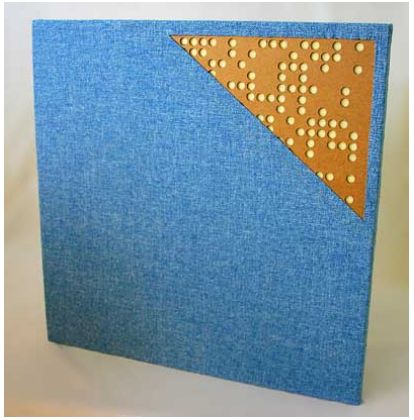
- Perforated panels:

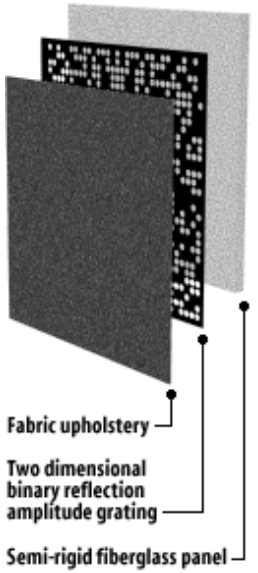
https://gikacoustics.net/



https://www.rpgeurope.com/products/product/bad.html





# Spaces with variable acoustics

Orchestral room ESMUC 293 m²

Sala 4 - Alicia de Larrocha
L'Auditori 165 m²

# Bibliography

QUIZ 1


BOOKS

- BALLOU, Glen. *Handbook for sound engineers.* Taylor & Francis, 2013.
- RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012.
- SELF, Douglas. *Audio engineering explained*. Routledge, 2012.


LINKS

- Matlab calculators: *http://www.mcsquared.com/java.htm*
- Sound. The wave equation: https://www.feynmanlectures.caltech.edu/I_47.html
- Sound Physics: https://www.britannica.com/science/sound-physics
- Basics on Music acoustics: http://www.phys.unsw.edu.au/jw/basics.html#string
- Calculators: http://www.sengpielaudio.com
- European Agency for Safety and Health at Work: https://oshwiki.osha.europa.eu/en/themes/noise
- Measuring Time Delay using Cross Correlation: https://www.youtube.com/watch?v=L6YJqhbsuFY
- Sound reinforcement: https://es.yamaha.com/es/products/contents/proaudio/docs/audio_quality/04_audio_quality.html
- Recording studio design: https://www.soundonsound.com/techniques/sos-guide-control-room-design
- Diffusor calculator: http://www.mh-audio.nl/Acoustics.html#top
- Acoustic panels and diffusors: https://www.rpgacoustic.com/

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